

CSE 311:

Data Communication

Instructor:
Dr. Md. Monirul Islam

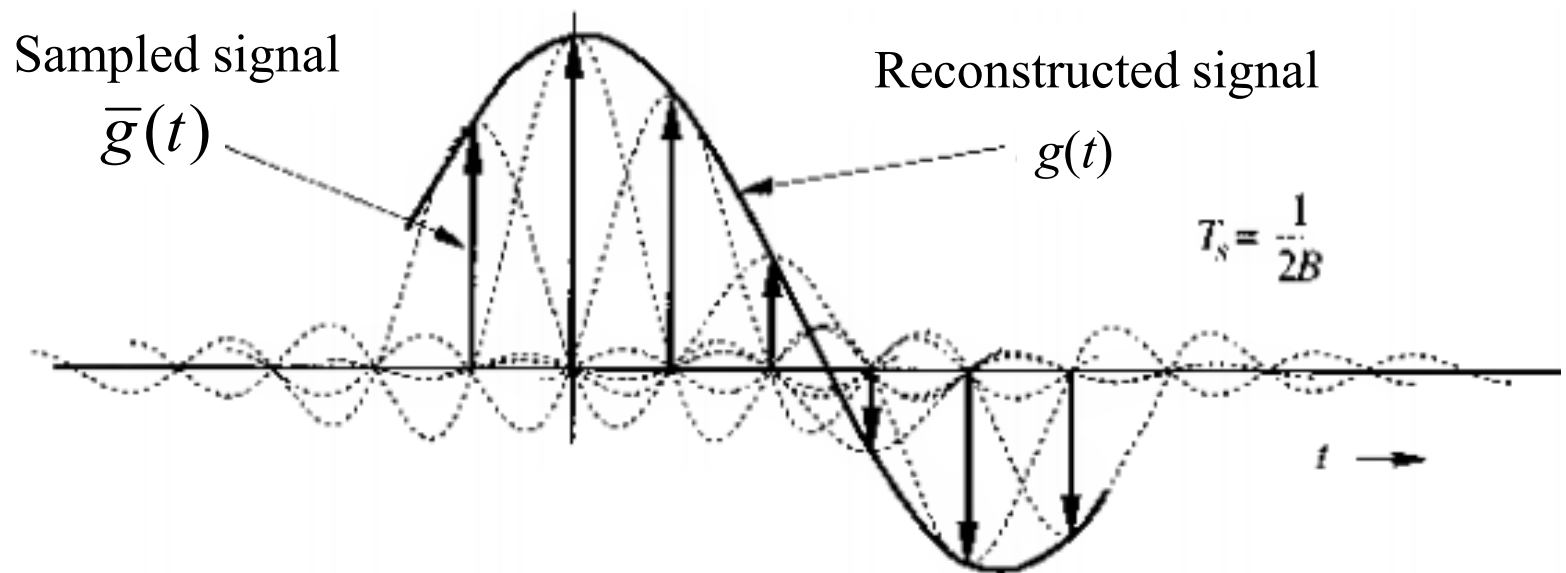
Sampling and Analog-to-Digital Conversion

Signal Reconstruction from Uniform Samples

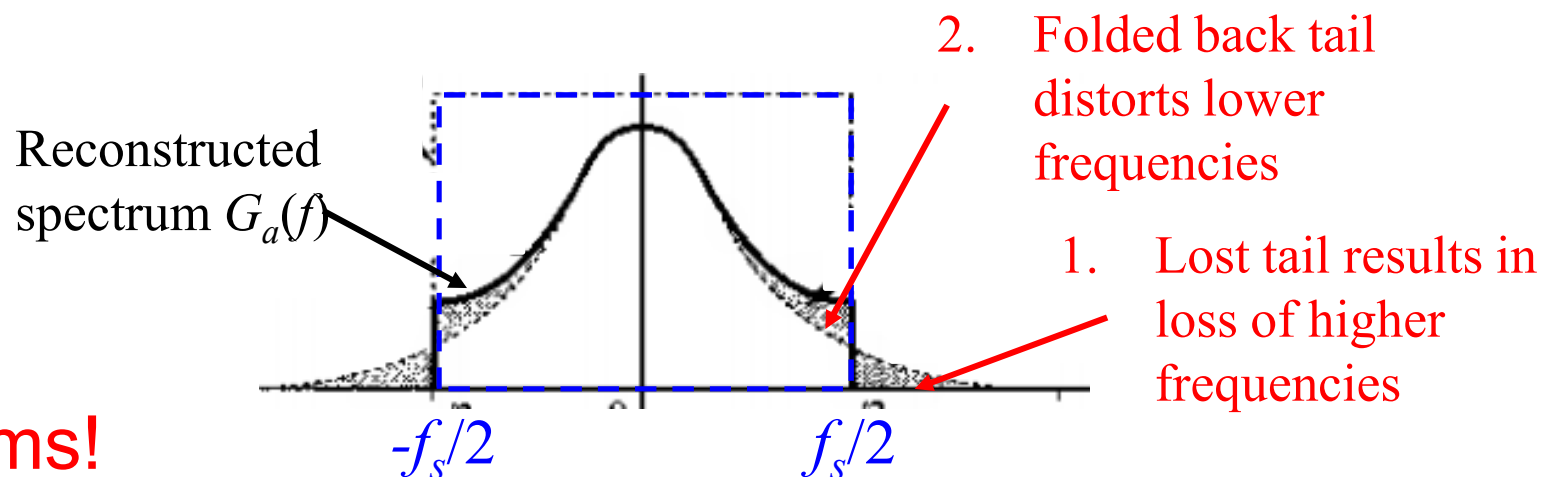
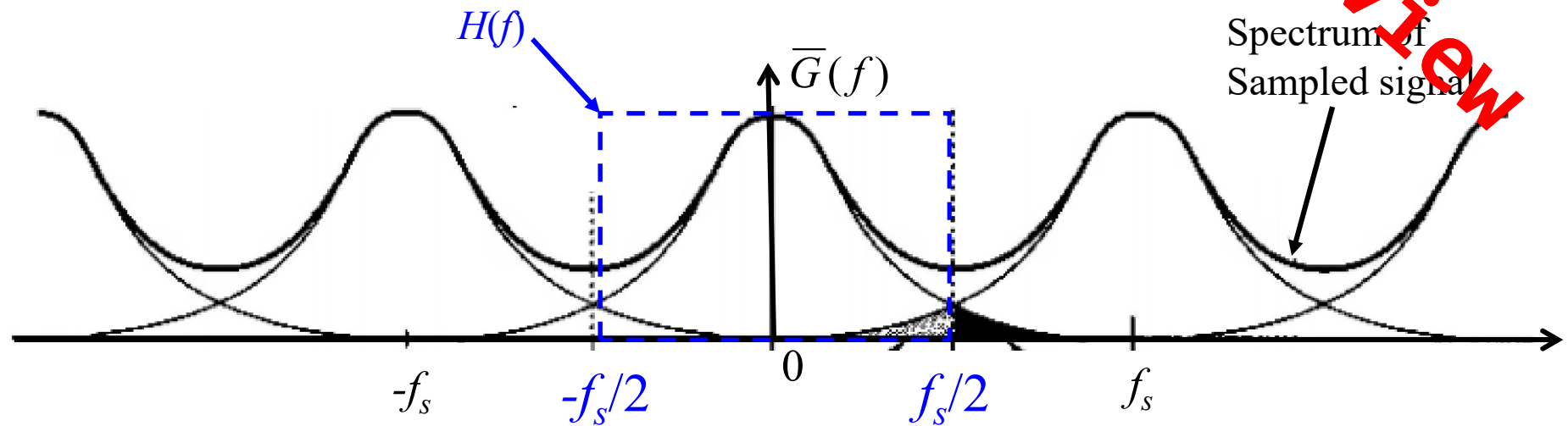
Review

Significance of Interpolation formula

$$g(t) = \sum_n g(nT_s) \text{sinc}(2\pi Bt - n\pi)$$

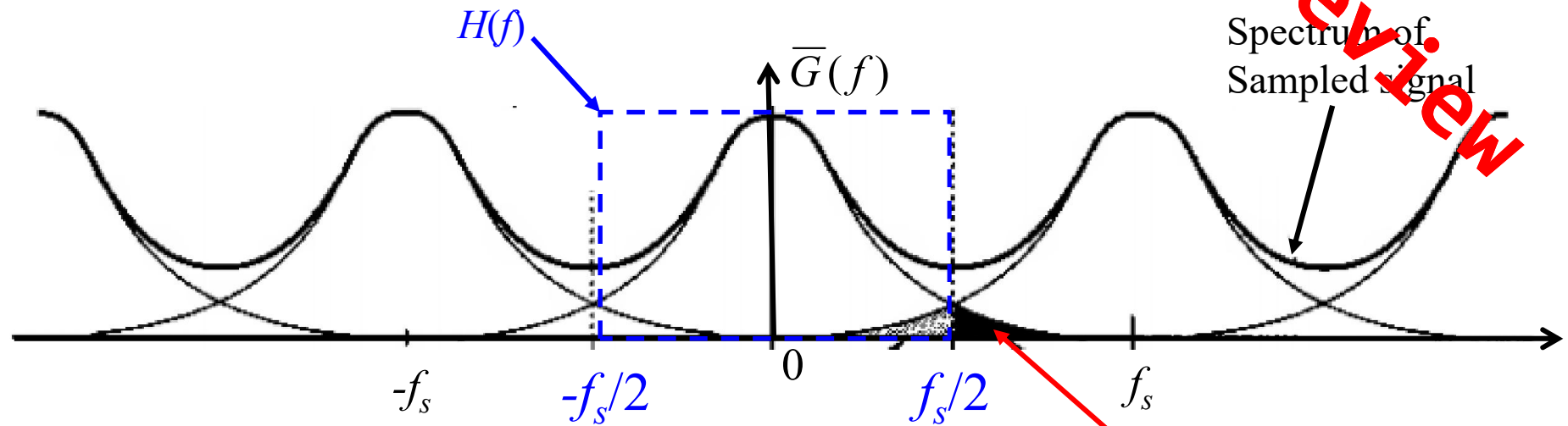


Wrong Assumption on Bandwidth: Treachery of Aliasing



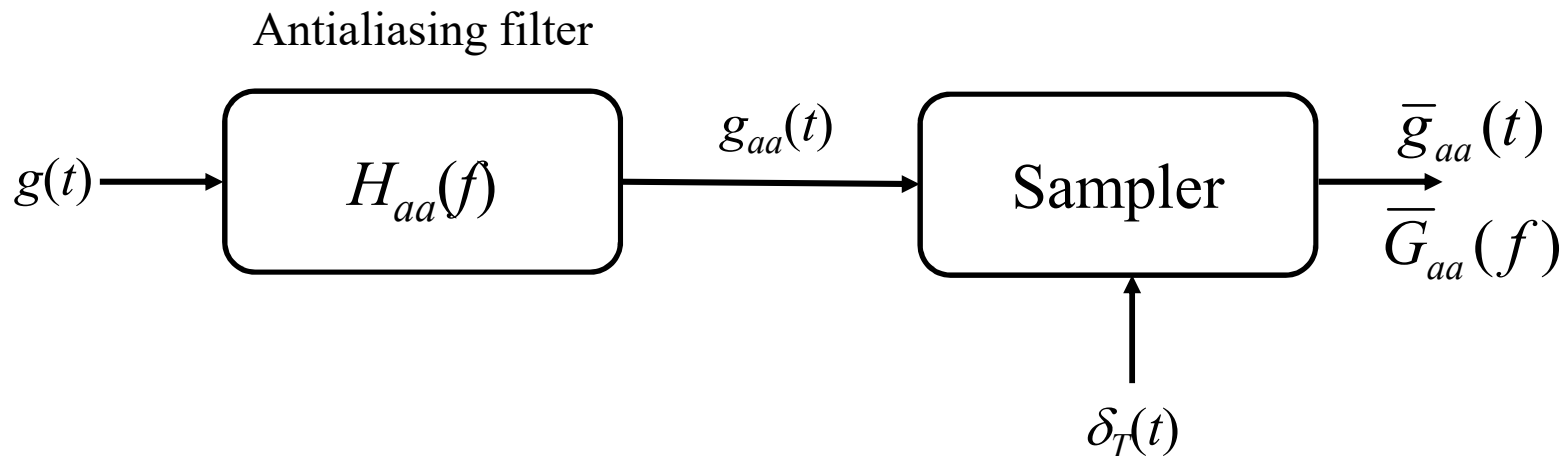
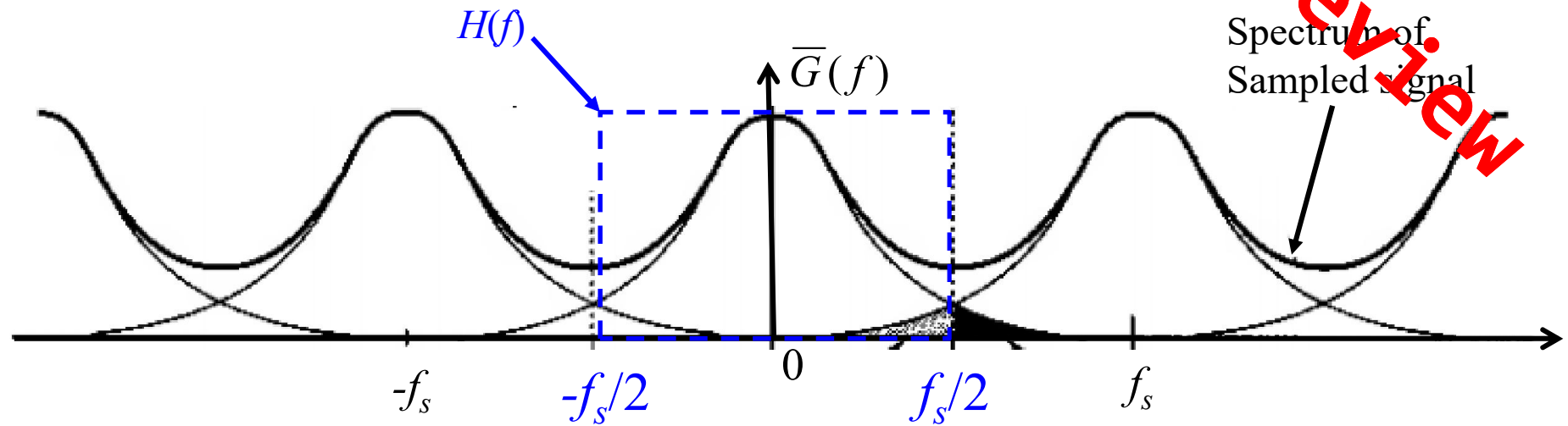
**TWO
Problems!**

Combating Aliasing

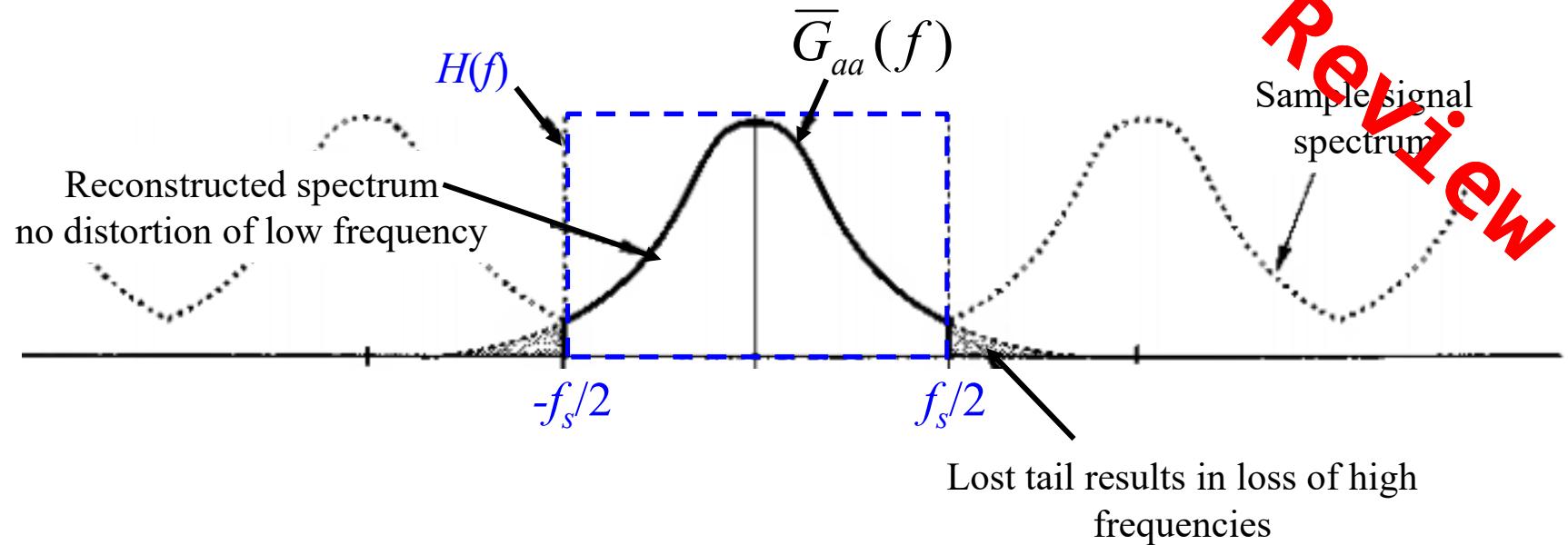


REMOVE this
portion before
sampling using an
Antialiasing filter
(ILPF)

Combating Aliasing



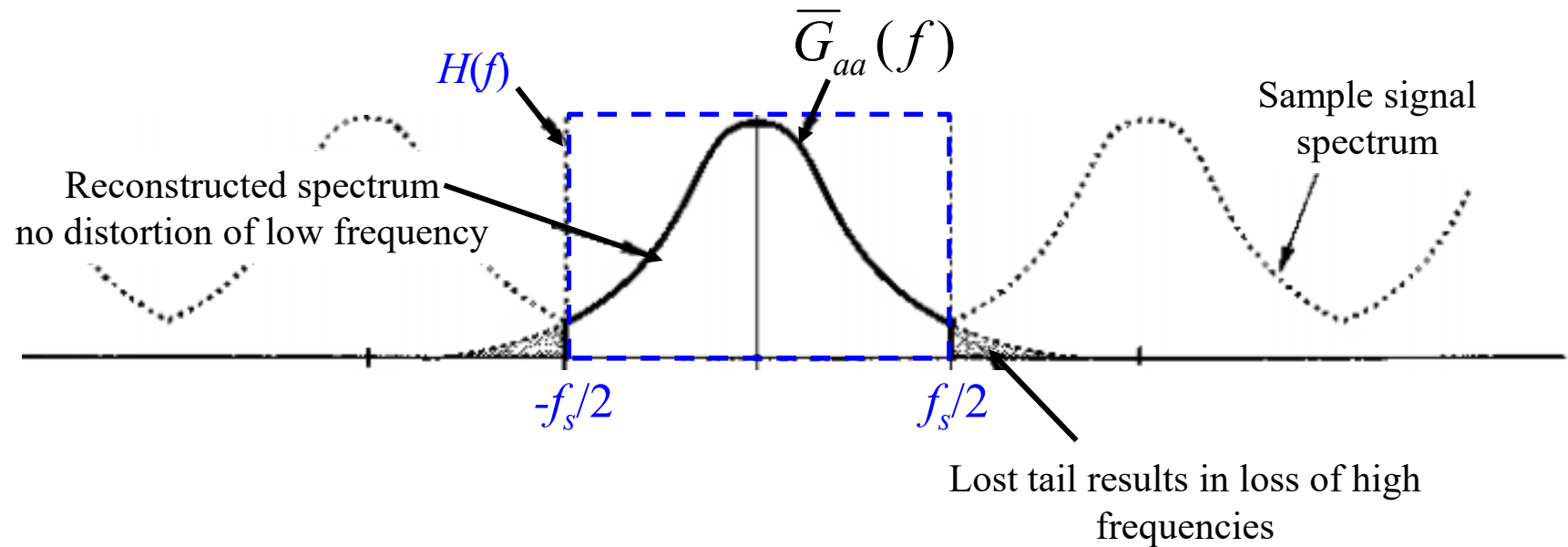
Combating Aliasing



$$\overline{G}_{aa}(f) = G(f) \quad |f| < f_s / 2$$

Advantage: suppressed
component cannot reappear
in low frequency
components

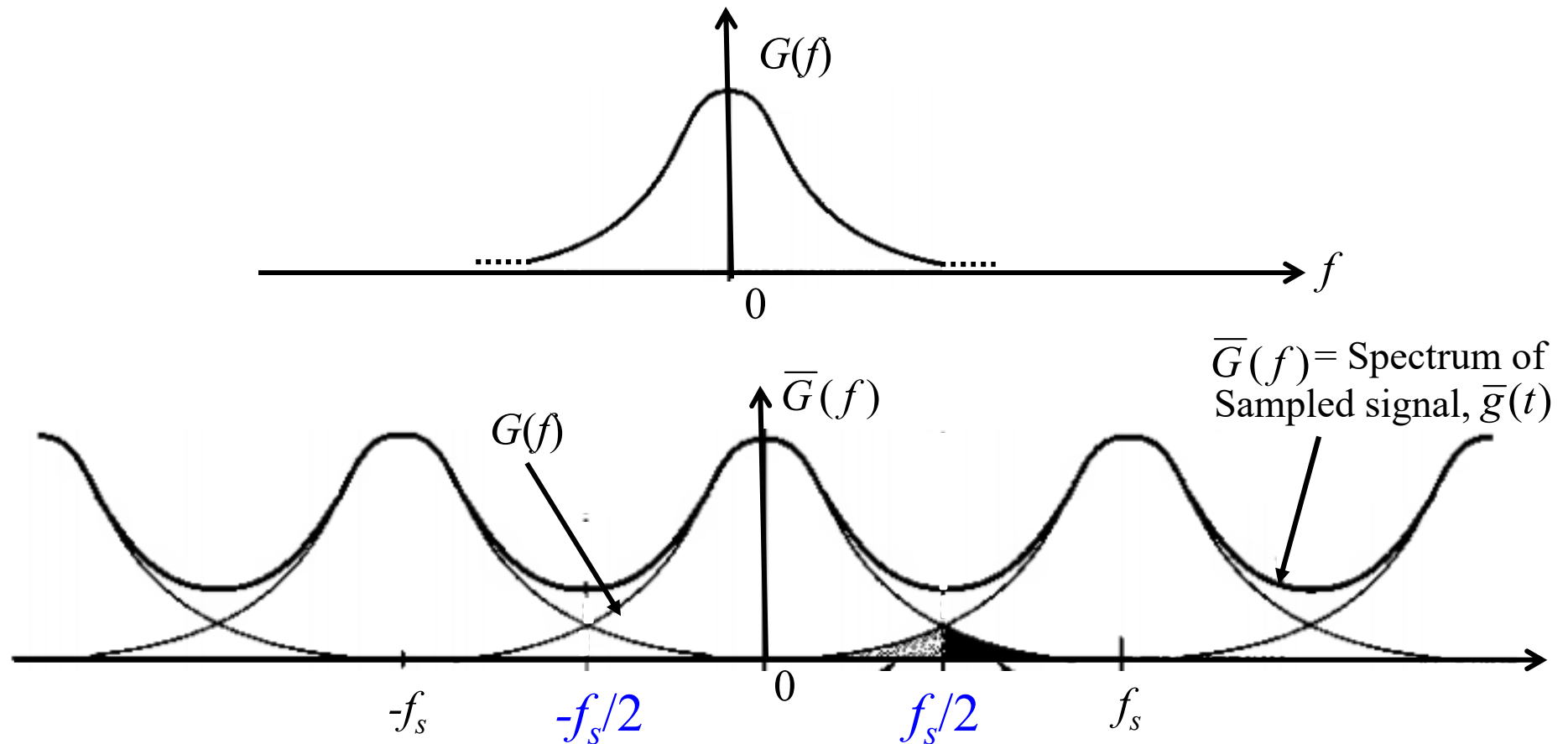
Combating Aliasing



Other characteristics:

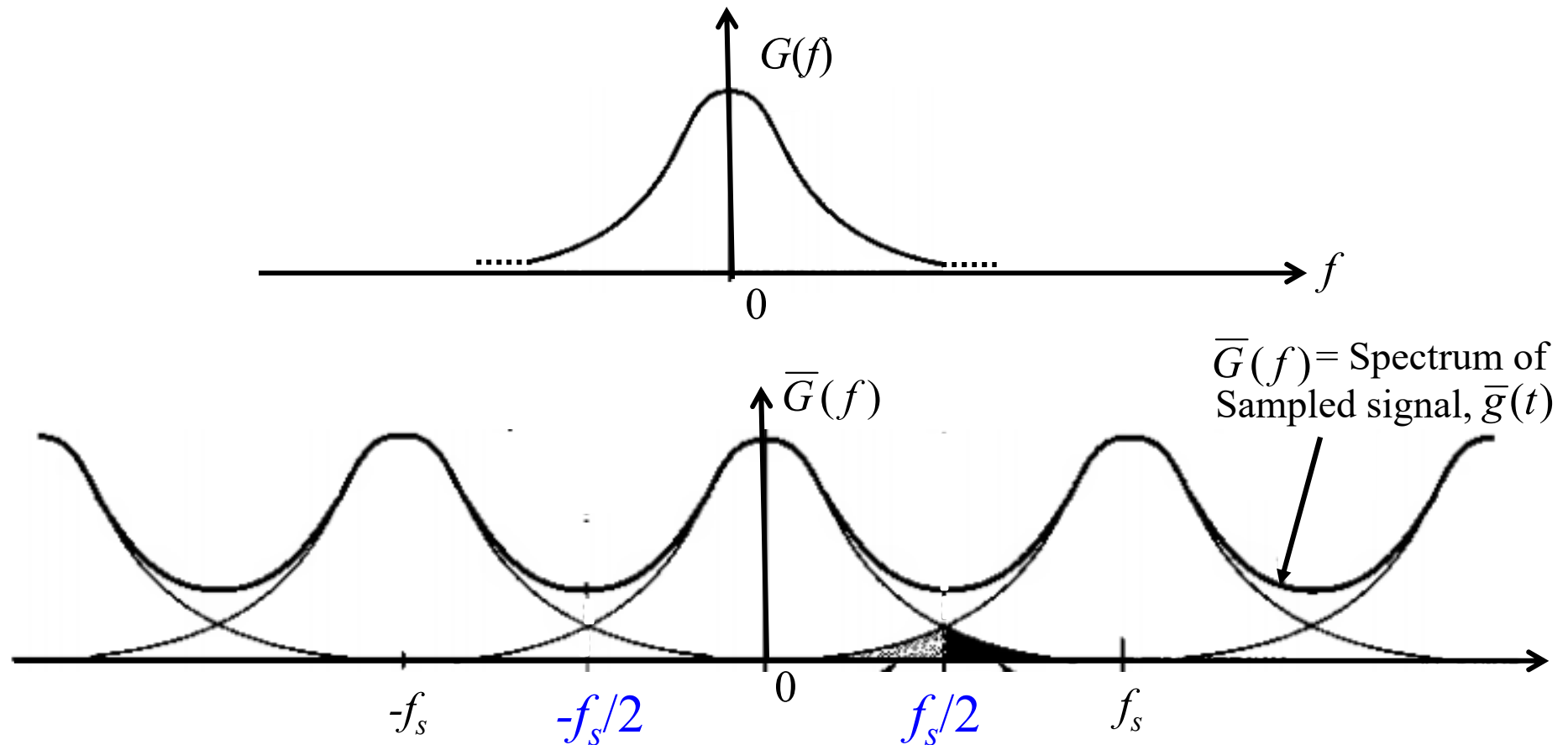
- Antialiasing filter band-limits $g(t)$ to $f_s/2$
- Reduces noise too, otherwise noise might be folded back to low frequency components
- Noises beyond $f_s/2$ are suppressed

Sampling Effect: Non-band-limited signal to band-limited signal



Bandwidth of $G(f)$ is practically infinite

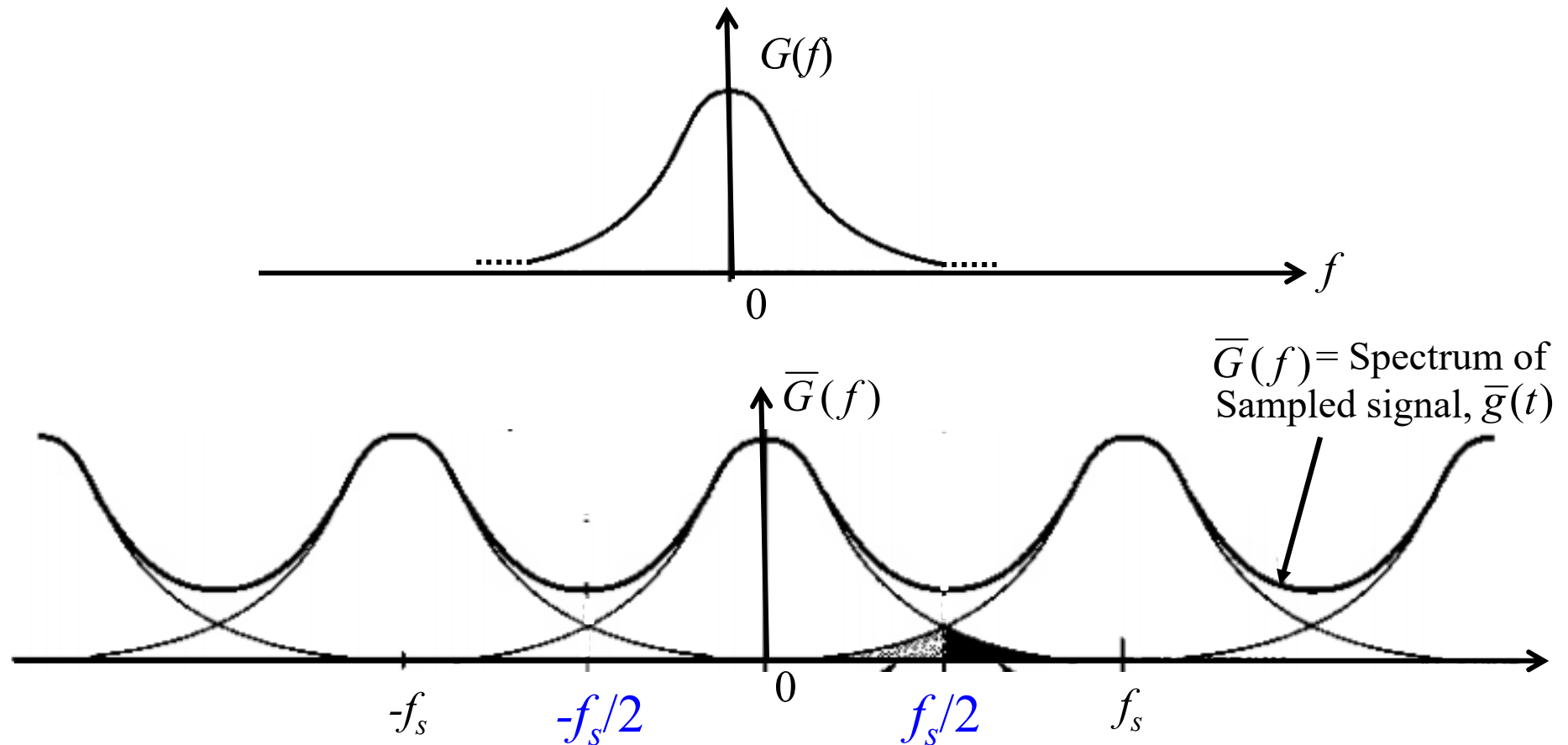
Sampling Effect: Non-band-limited signal to band-limited signal



Bandwidth of $G(f)$ is practically infinite

$f_s \ll \text{Nyquist rate}$

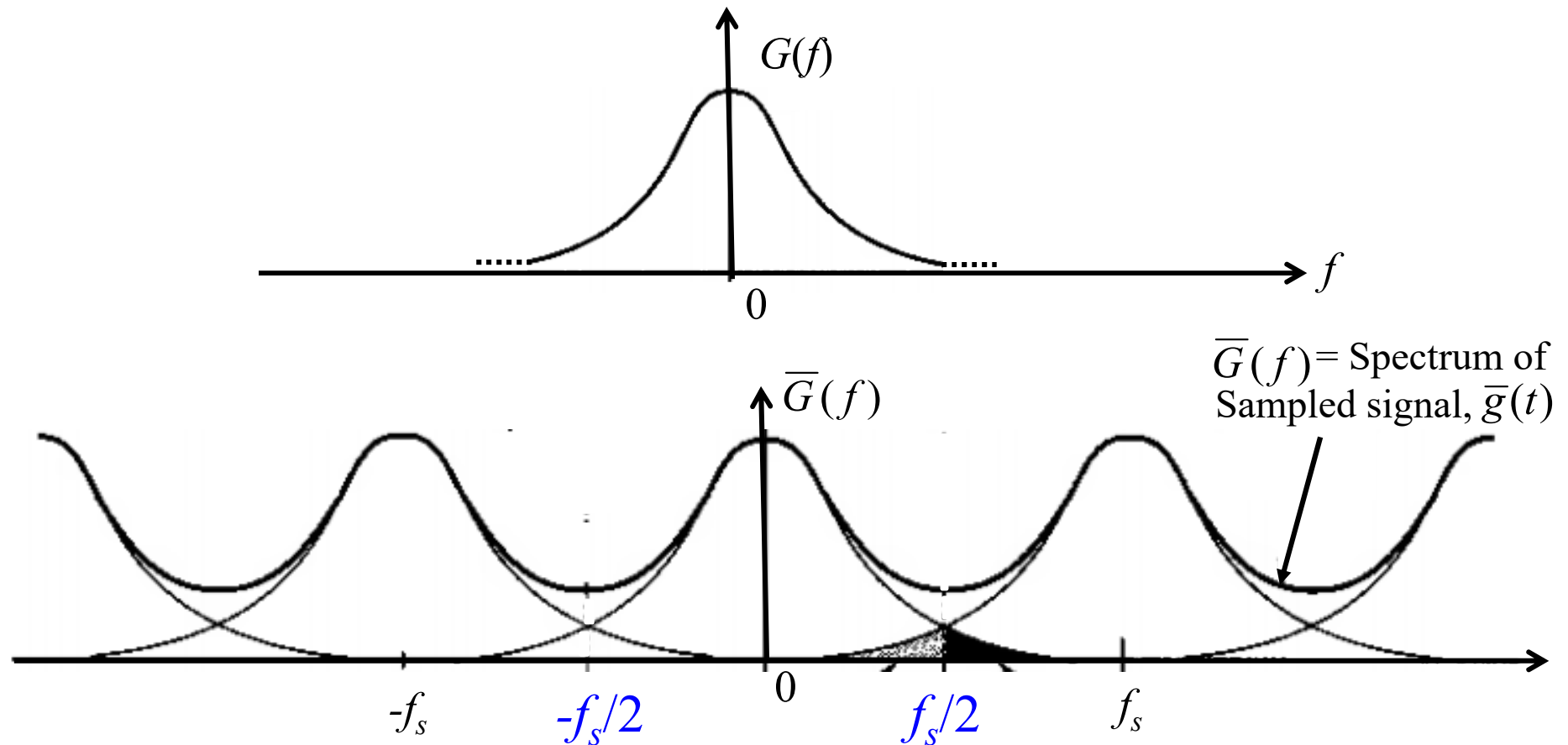
Sampling Effect: Non-band-limited signal to band-limited signal



Bandwidth of $G(f)$ is practically infinite

$\bar{G}(f)$, that means, spectrum of $\bar{g}(t)$ consists of overlapping cycles of $G(f)$

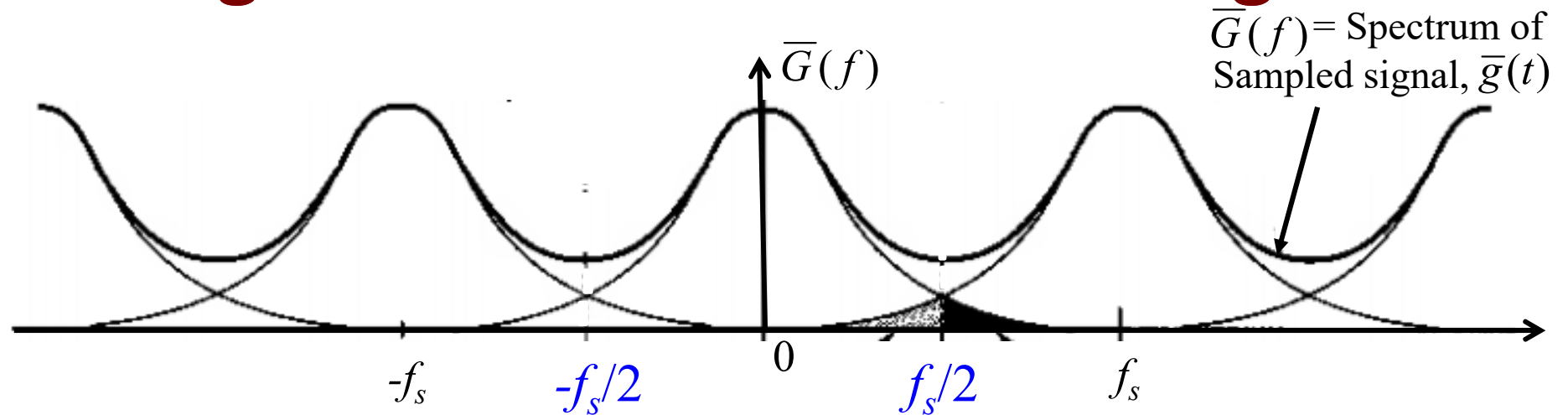
Sampling Effect: Non-band-limited signal to band-limited signal



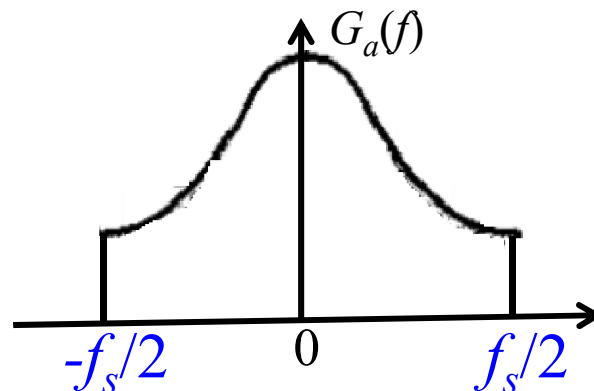
Bandwidth of $G(f)$ is practically infinite

$\bar{g}(t)$ are sub-Nyquist samples of $g(t)$!

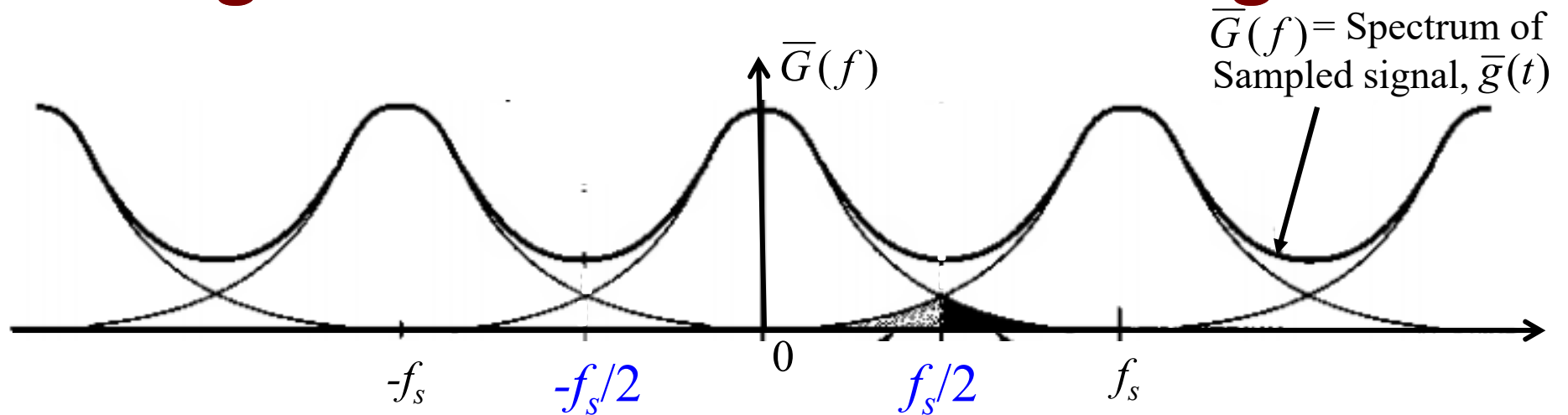
Sampling Effect: Non-band-limited signal to band-limited signal



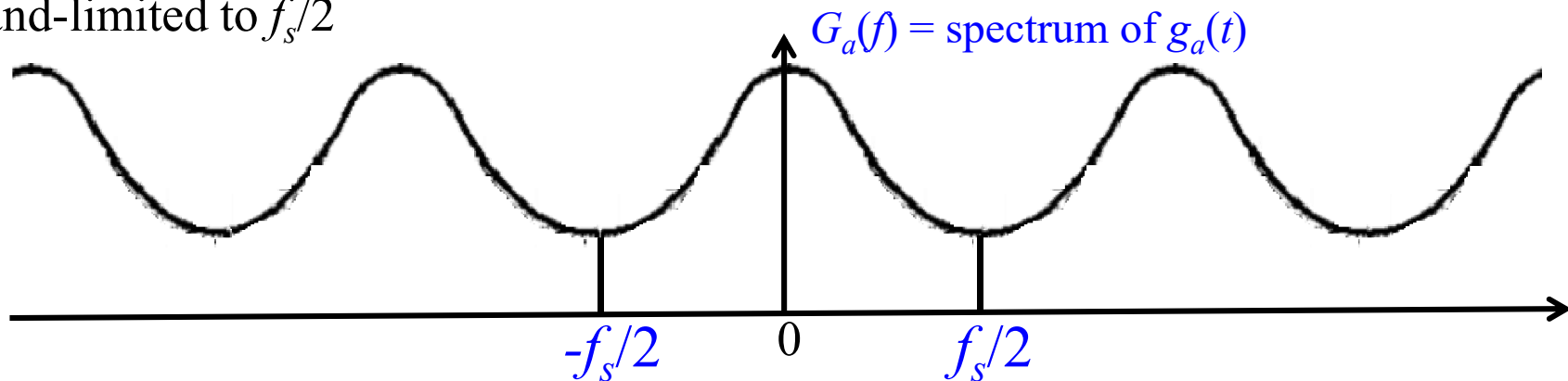
The **above bold line spectrum** can be viewed as **repetition** of the following $G_a(f)$ band-limited to $f_s/2$



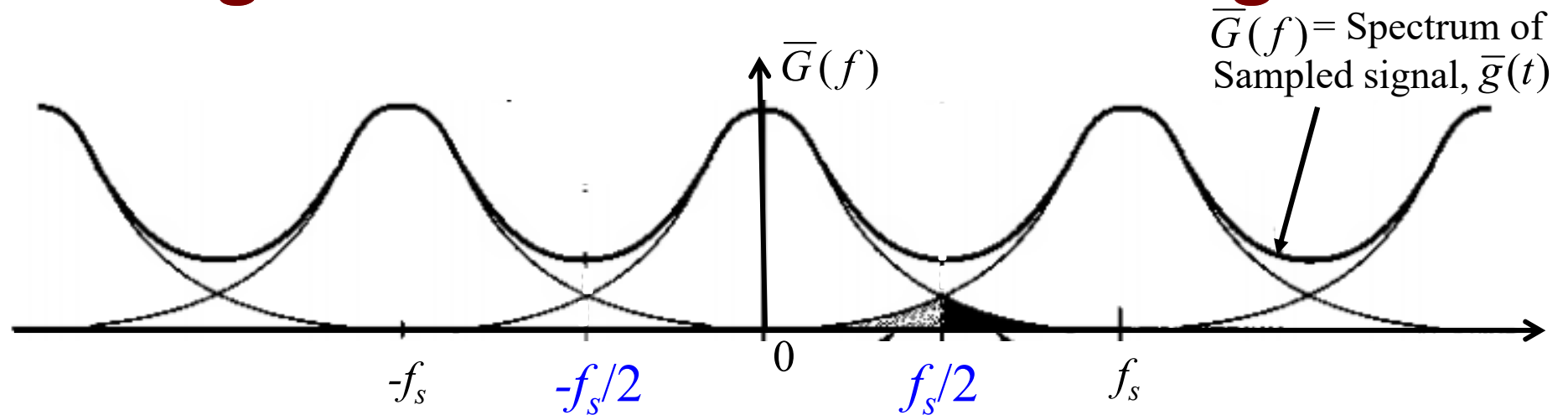
Sampling Effect: Non-band-limited signal to band-limited signal



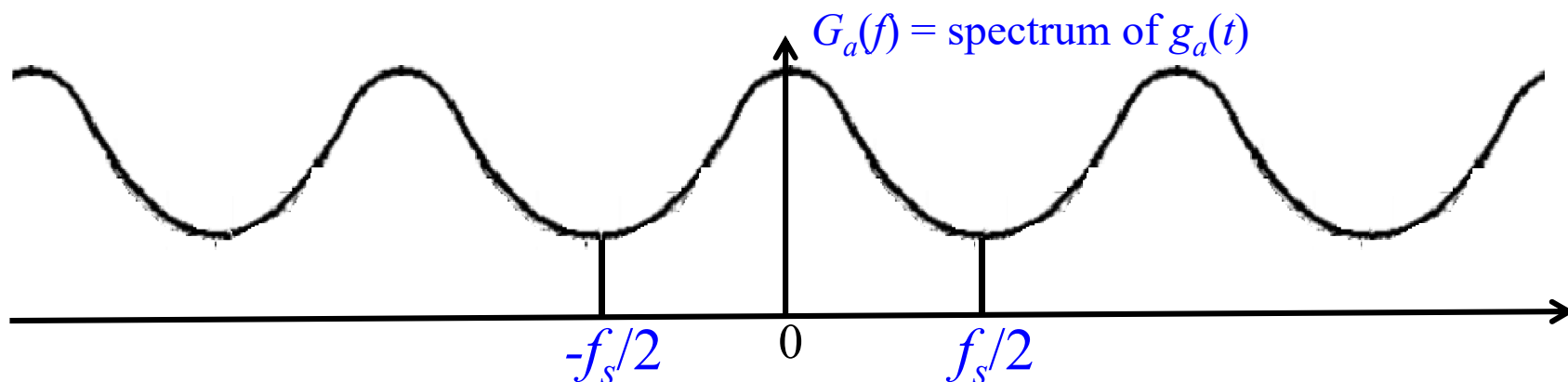
The above **bold line** spectrum can be viewed as repetition of the following $G_a(f)$ band-limited to $f_s/2$



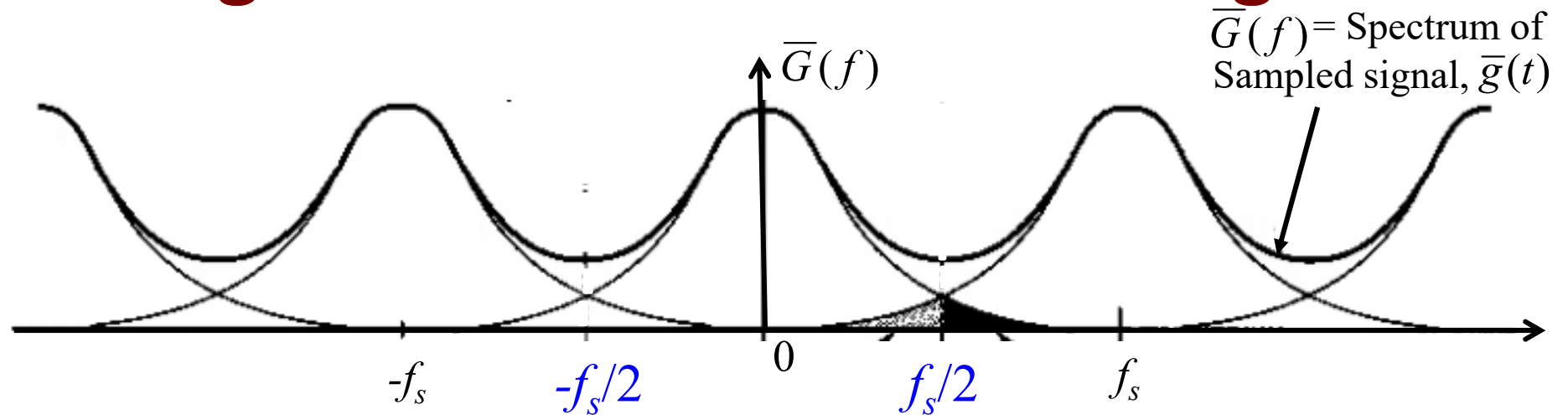
Sampling Effect: Non-band-limited signal to band-limited signal



$\bar{g}(t)$ are Nyquist samples of $g_a(t)$

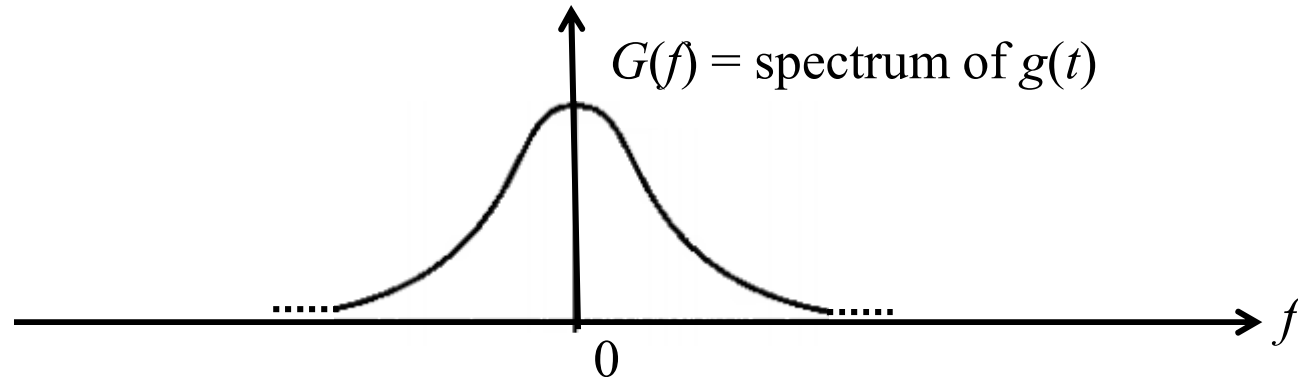


Sampling Effect: Non-band-limited signal to band-limited signal



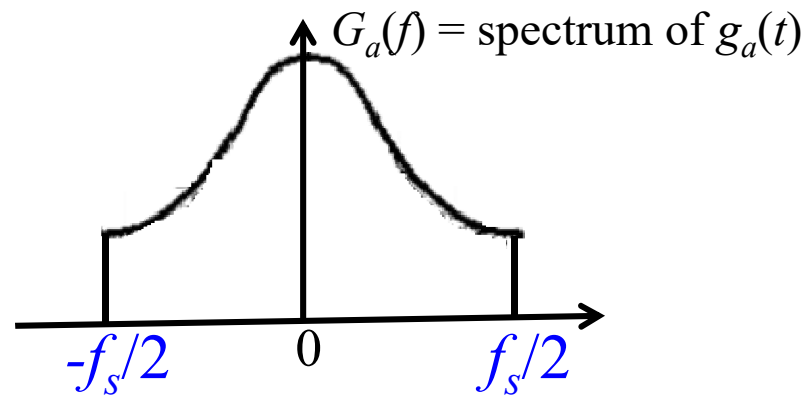
Sampling a non-band-limited signal $g(t)$ at f_s is equivalent to Nyquist sampling of some signal $g_a(t)$ band-limited to $f_s/2$

Sampling Effect: Non-band-limited signal to band-limited signal



Let
sub-Nyquist Sampling of $g(t)$ at f_s generates samples $g(0), g(T_s), g(2T_s), g(3T_s), \dots$

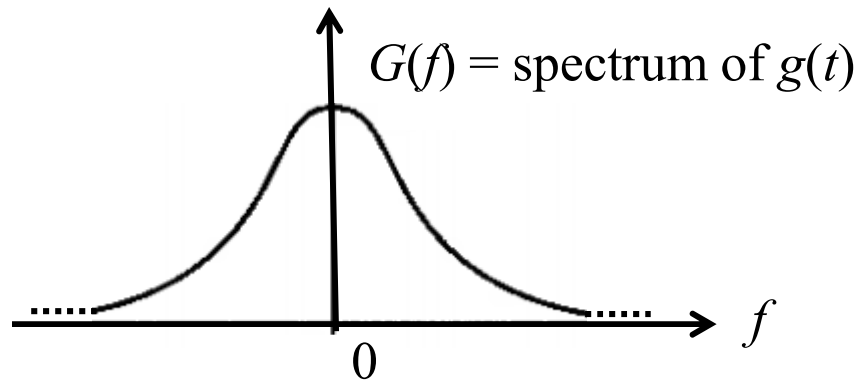
Sampling Effect: Non-band-limited signal to band-limited signal



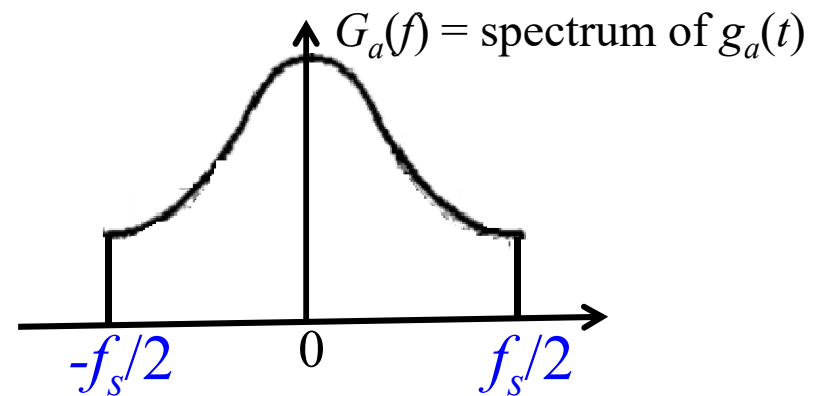
Let

Nyquist Sampling of $g_a(t)$ at f_s generates samples $g_a(0), g_a(T_s), g_a(2T_s), g_a(3T_s), \dots$

Sampling Effect: Non-band-limited signal to band-limited signal

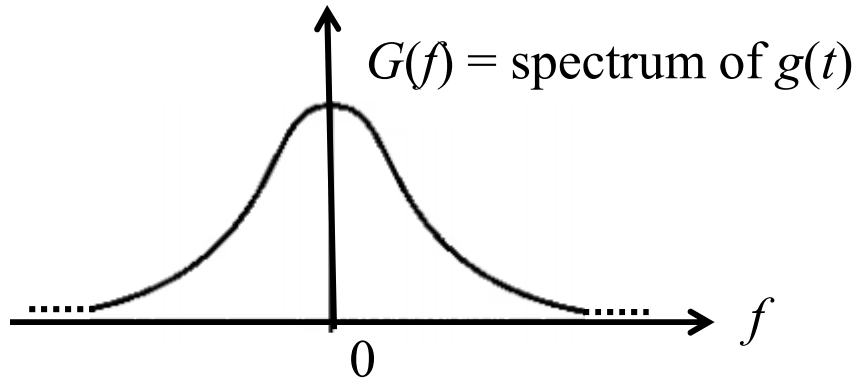


sub-Nyquist Samples $g(0)$,
 $g(T_s)$, $g(2T_s)$, $g(3T_s)$, $\dots$

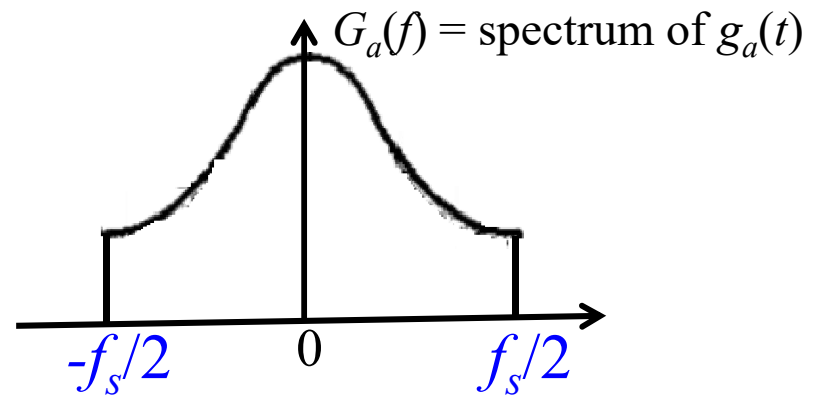


Nyquist Samples $g_a(0)$, $g_a(T_s)$, $g_a(2T_s)$,
 $g_a(3T_s)$, $\dots$

Sampling Effect: Non-band-limited signal to band-limited signal



sub-Nyquist Samples $g(0)$,
 $g(T_s)$, $g(2T_s)$, $g(3T_s)$, $\dots$

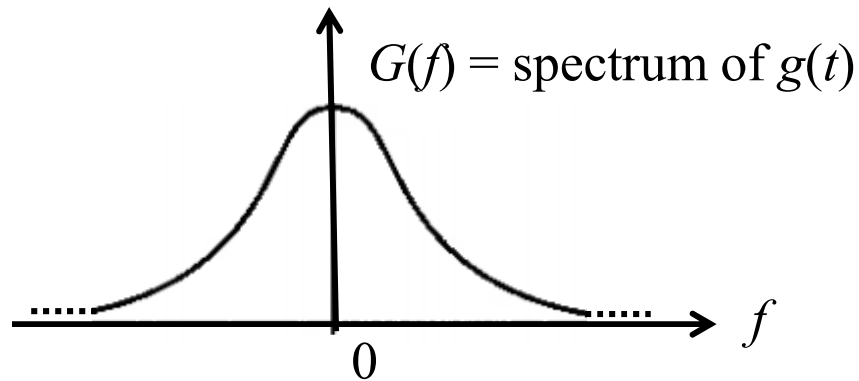


Nyquist Samples $g_a(0)$, $g_a(T_s)$, $g_a(2T_s)$,
 $g_a(3T_s)$, $\dots$

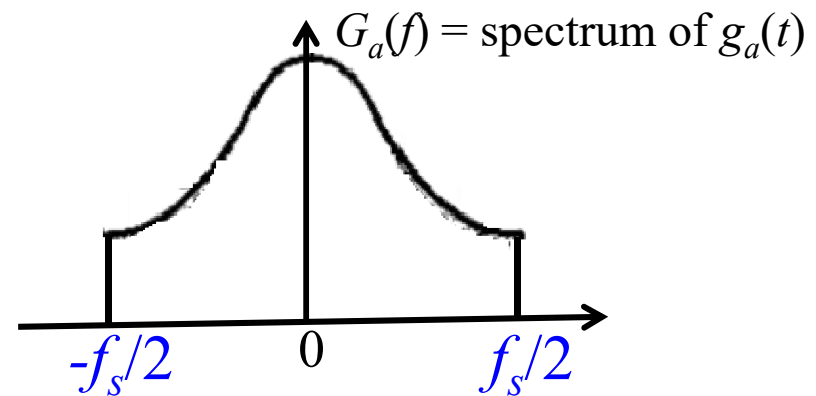
According to sampling effect that we saw,

$$g(0) = g_a(0), g(T_s) = g_a(T_s), g(2T_s) = g_a(2T_s), g(3T_s) = g_a(3T_s), \text{ and so on}$$

Sampling Effect: Non-band-limited signal to band-limited signal



sub-Nyquist Samples $g(0)$,
 $g(T_s)$, $g(2T_s)$, $g(3T_s)$, $\dots$

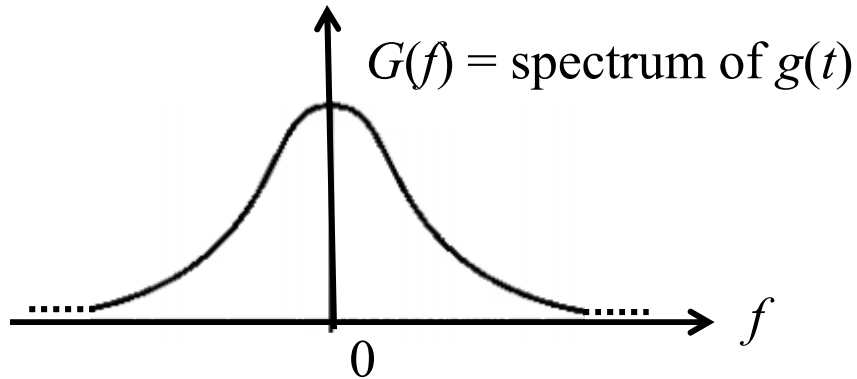


Nyquist Samples $g_a(0)$, $g_a(T_s)$, $g_a(2T_s)$,
 $g_a(3T_s)$, $\dots$

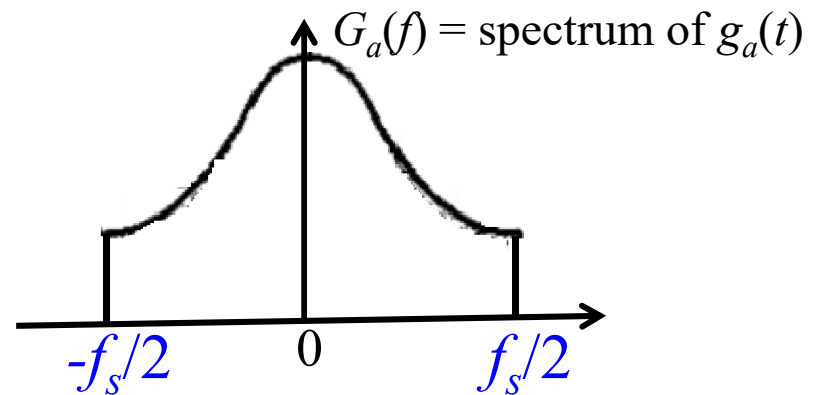
Therefore,

$$g(nT_s) = g_a(nT_s) = g_n$$

Sampling Effect: Non-band-limited signal to band-limited signal



sub-Nyquist Samples $g(0)$,
 $g(T_s)$, $g(2T_s)$, $g(3T_s)$, $\dots$



Nyquist Samples $g_a(0)$, $g_a(T_s)$, $g_a(2T_s)$,
 $g_a(3T_s)$, $\dots$

Therefore,

$$g(nT_s) = g_a(nT_s) = g_n$$

In other words, sampling $g(t)$ and $g_a(t)$ at the rate of $f_s = 1/T_s$ will generate the **same independent** data sequence

Maximum Information Rate of a Channel with BW = B Hz

Assume

- Error free, noise less channel
- Channel bandwidth is B

Maximum Information Rate of a Channel with BW = B Hz

Assume

- Error free, noise less channel
- Channel bandwidth is B

Information rate

- Maximum $2B$ pieces of information can be sent per second

Maximum Information Rate of a Channel with BW = B Hz

Assume

- Error free, noise less channel
- Channel bandwidth is B

Information rate

- Channel can send a low pass signal of B Hz
- The signal can be recovered from samples uniformly taken at $2B$ samples per second

Maximum Information Rate of a Channel with BW = B Hz

Assume

- Error free, noise less channel
- Channel bandwidth is B

Information rate

- Channel can send a low pass signal of B Hz
- The signal can be recovered from samples uniformly taken at $2B$ samples per second
- This means, $2B$ samples/second can be sent through the channel

Maximum Information Rate of a Channel with BW = B Hz

Assume

- Error free, noise less channel
- Channel bandwidth is B

Information rate

- Channel can send a low pass signal of B Hz
- The signal can be recovered from samples uniformly taken at $2B$ samples per second
- This means, $2B$ samples/second can be sent through the channel

We have to prove that

- A sequence of data samples at the rate of $2B$ samples/second can come from uniform sampling of a signal of bandwidth B Hz
- The signal can be recovered from these samples

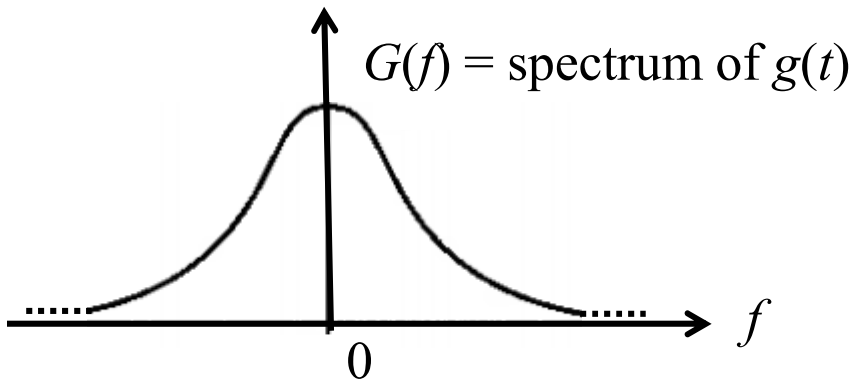
Maximum Information Rate of a Channel with BW = B Hz

Assume a sequence of samples $g_0, g_1, g_2, g_3, \dots$ denoted as $\{g_n\}$ at the rate of $2B$ s/s

Maximum Information Rate of a Channel with BW = B Hz

Assume a sequence of samples $g_0, g_1, g_2, g_3, \dots$ denoted as $\{g_n\}$ at the rate of $2B$ s/s

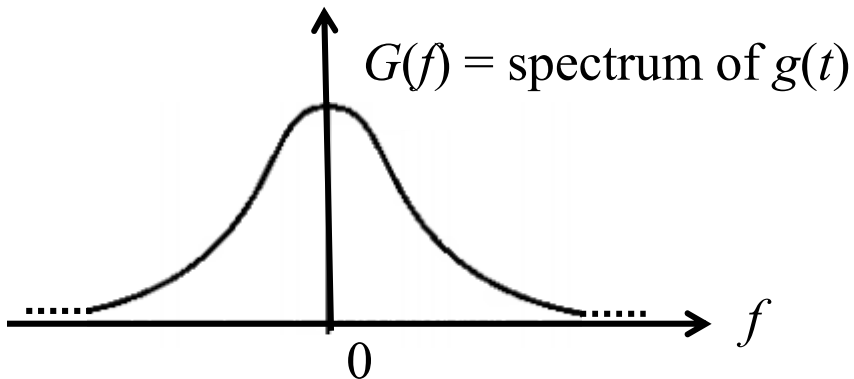
We will always find a signal $g(t)$ whose samples $g(0), g(T_s), g(2T_s), g(3T_s), \dots$ matches with $\{g_n\}$.



Maximum Information Rate of a Channel with BW = B Hz

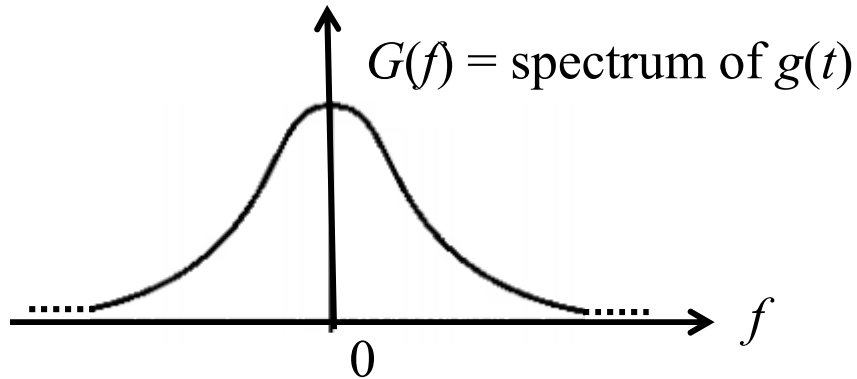
Assume a sequence of samples $g_0, g_1, g_2, g_3, \dots$ denoted as $\{g_n\}$ at the rate of $2B$ s/s

We will always find a signal $g(t)$ whose samples $g(0), g(T_s), g(2T_s), g(3T_s), \dots$ matches with $\{g_n\}$.

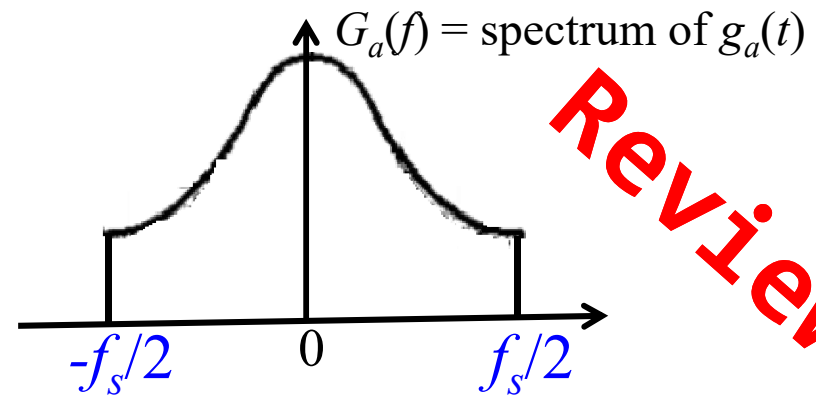


This means,
 $g_n = g(nT_s)$

Sampling Effect: Non-band-limited signal to band-limited signal



sub-Nyquist Samples $g(0)$,
 $g(T_s)$, $g(2T_s)$, $g(3T_s)$, $\dots$



Review

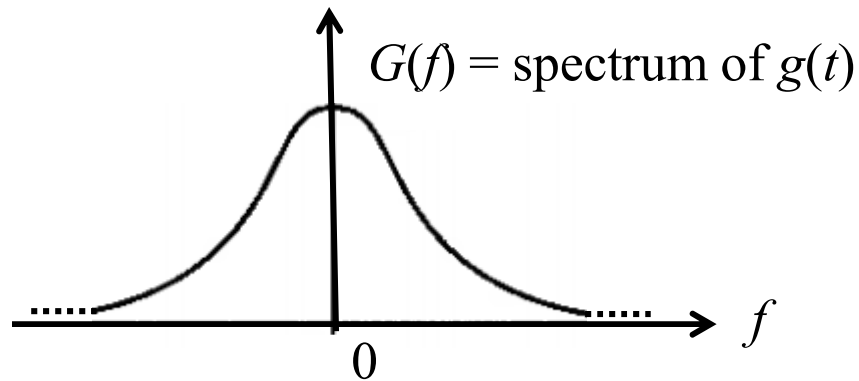
Nyquist Samples $g_a(0)$, $g_a(T_s)$, $g_a(2T_s)$,
 $g_a(3T_s)$, $\dots$

Therefore,

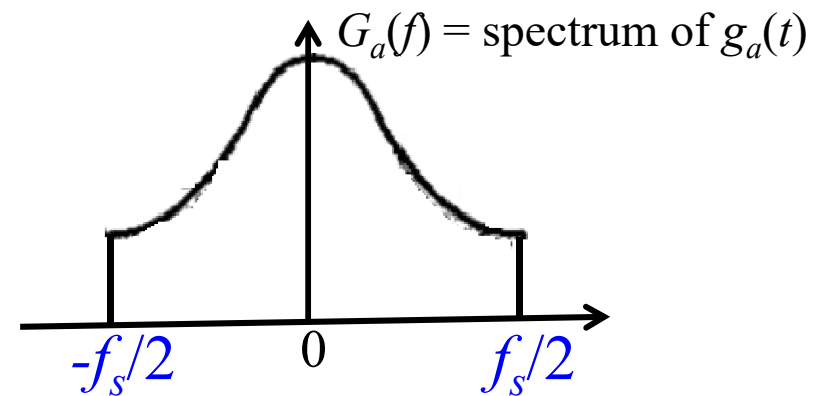
$$g(nT_s) = g_a(nT_s) = g_n$$

In other words, sampling $g(t)$ and $g_a(t)$ at the rate of $f_s = 1/T_s$ will generate the same independent data sequence

Sampling Effect: Non-band-limited signal to band-limited signal



sub-Nyquist Samples $g(0)$,
 $g(T_s)$, $g(2T_s)$, $g(3T_s)$, $\dots$

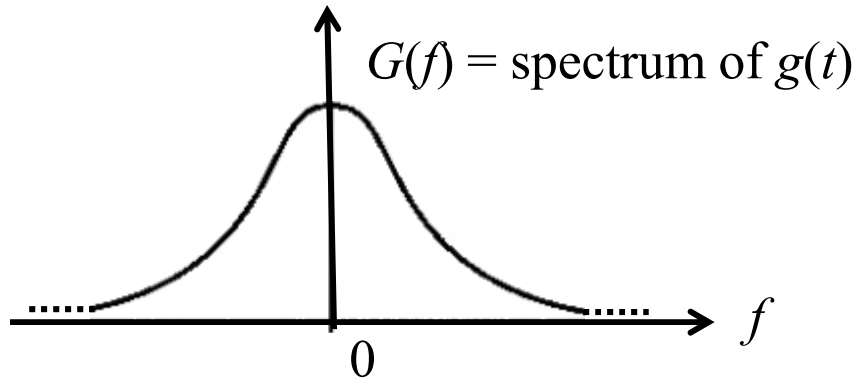


Nyquist Samples $g_a(0)$, $g_a(T_s)$, $g_a(2T_s)$,
 $g_a(3T_s)$, $\dots$

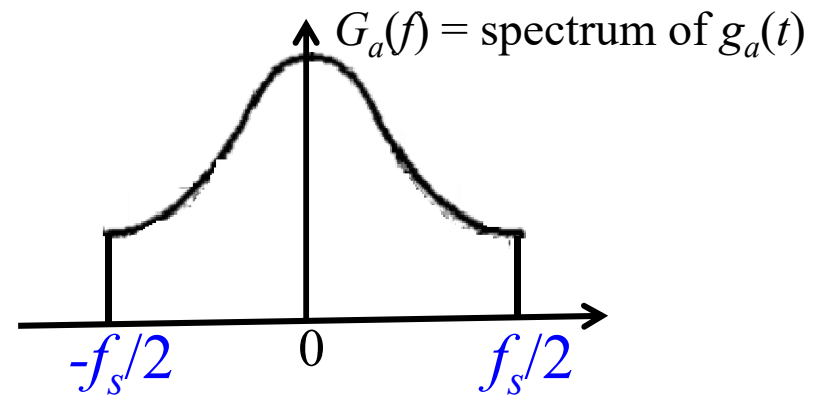


$$\bar{g}(t) = \sum_n g(nT_s) \delta(t - nT_s)$$

Sampling Effect: Non-band-limited signal to band-limited signal

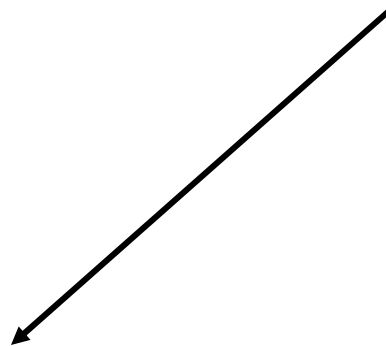


sub-Nyquist Samples $g(0)$,
 $g(T_s)$, $g(2T_s)$, $g(3T_s)$, $\dots$

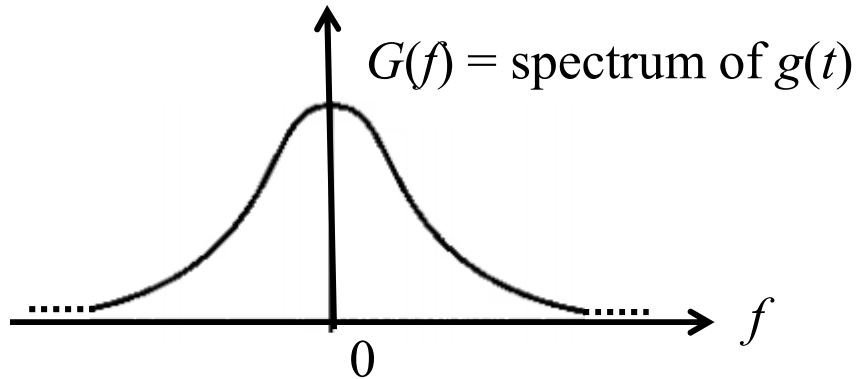


Nyquist Samples $g_a(0)$, $g_a(T_s)$, $g_a(2T_s)$,
 $g_a(3T_s)$, $\dots$

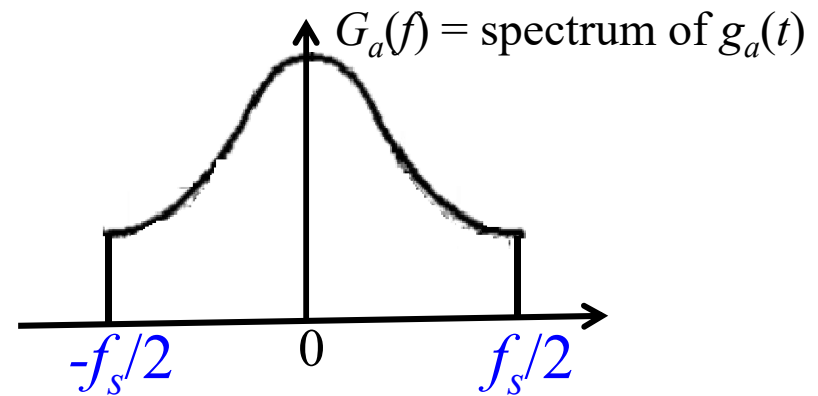
$$\begin{aligned}\bar{g}(t) &= \sum_n g(nT_s) \delta(t - nT_s) \\ &= \sum_n g_a(nT_s) \delta(t - nT_s)\end{aligned}$$



Sampling Effect: Non-band-limited signal to band-limited signal



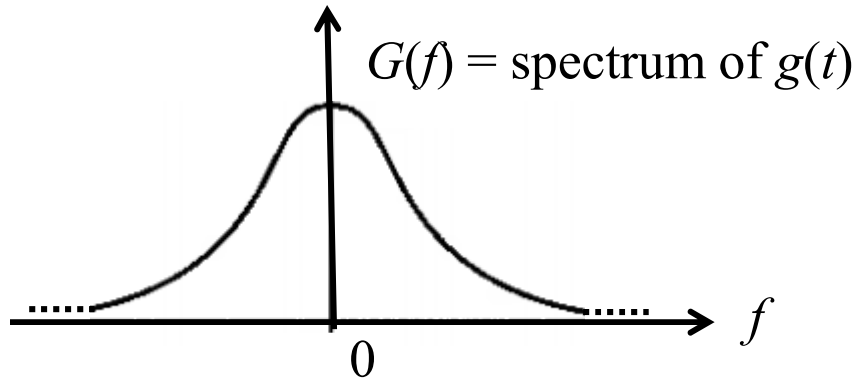
sub-Nyquist Samples $g(0)$,
 $g(T_s)$, $g(2T_s)$, $g(3T_s)$, $\dots$



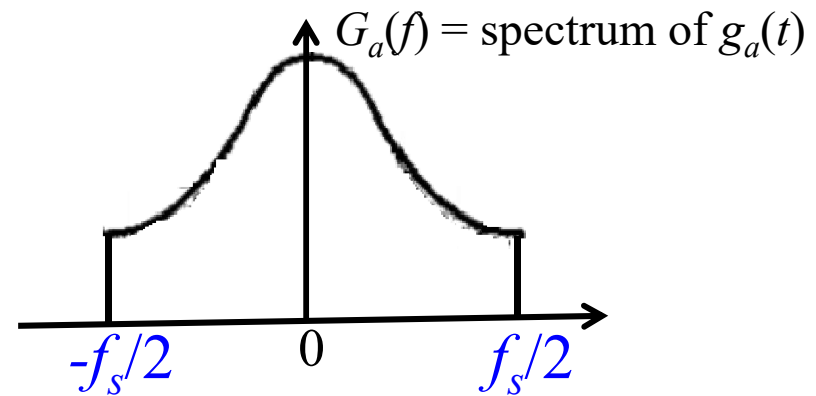
Nyquist Samples $g_a(0)$, $g_a(T_s)$, $g_a(2T_s)$,
 $g_a(3T_s)$, $\dots$

$$\begin{aligned}\bar{g}(t) &= \sum_n g(nT_s) \delta(t - nT_s) \\ &= \sum_n g_a(nT_s) \delta(t - nT_s) \\ &= \sum_n g_n \delta(t - nT_s)\end{aligned}$$

Sampling Effect: Non-band-limited signal to band-limited signal



sub-Nyquist Samples $g(0)$,
 $g(T_s)$, $g(2T_s)$, $g(3T_s)$, ...



Nyquist Samples $g_a(0)$, $g_a(T_s)$, $g_a(2T_s)$,
 $g_a(3T_s)$, ...

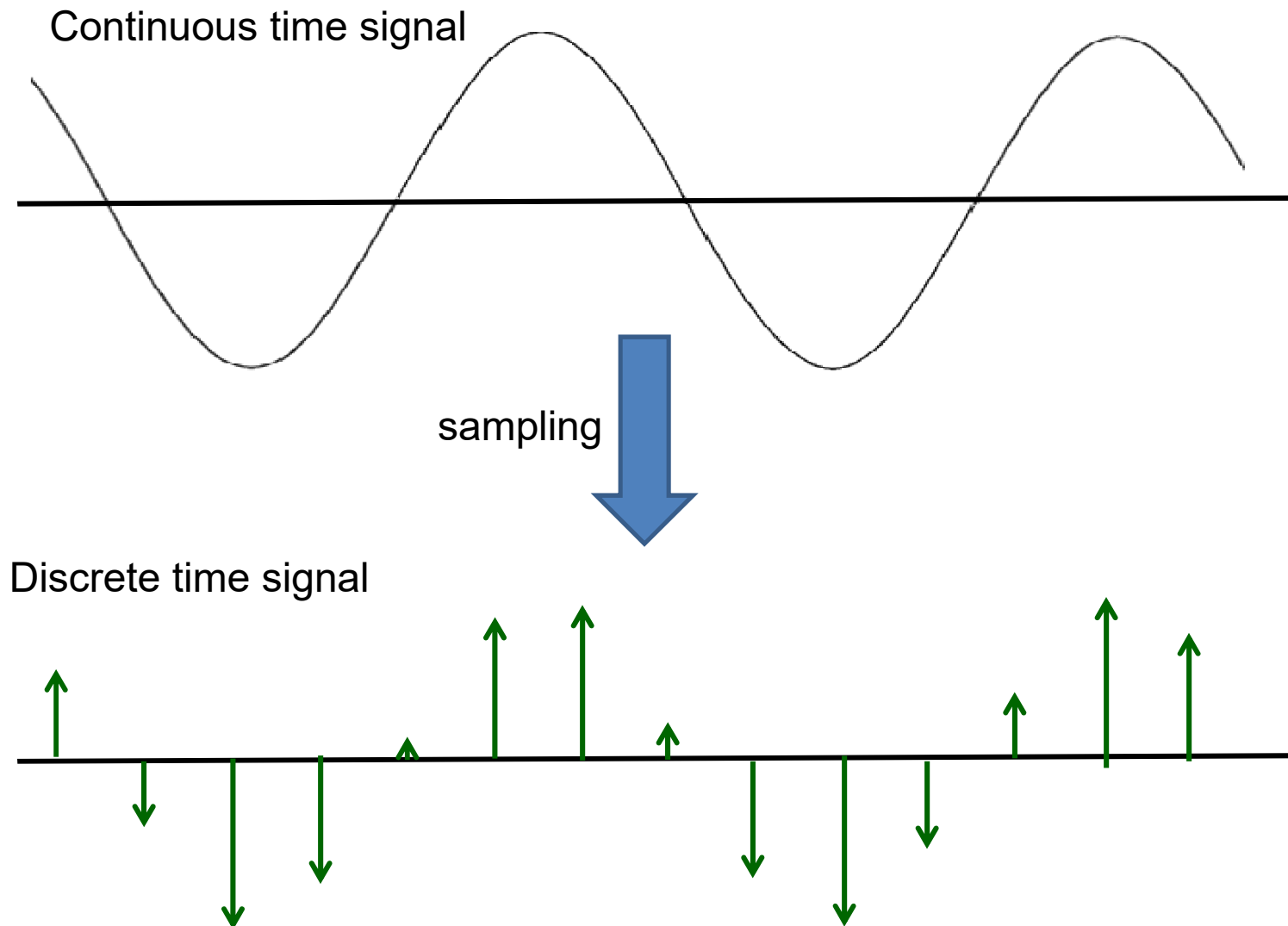
↓

$$\begin{aligned}\bar{g}(t) &= \sum_n g(nT_s) \delta(t - nT_s) \\ &= \sum_n g_a(nT_s) \delta(t - nT_s) \\ &= \sum_n g_n \delta(t - nT_s)\end{aligned}$$

To recover $g_a(t)$, we can use $\{g_n\}$ using,

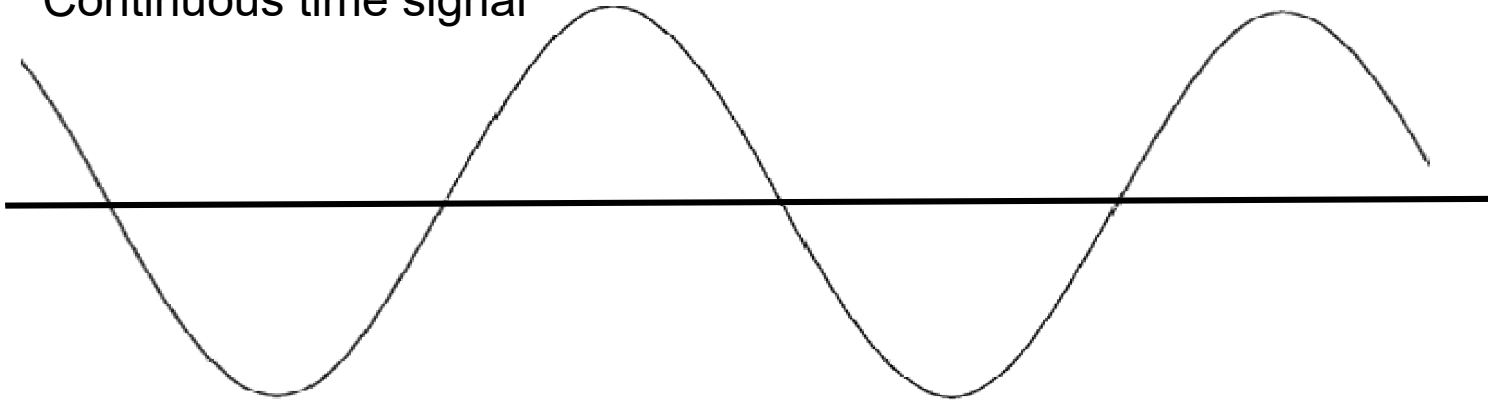
$$g_a(t) = \sum_n g_n \text{sinc}(2\pi Bt - n\pi)$$

Application of Sampling Theorem

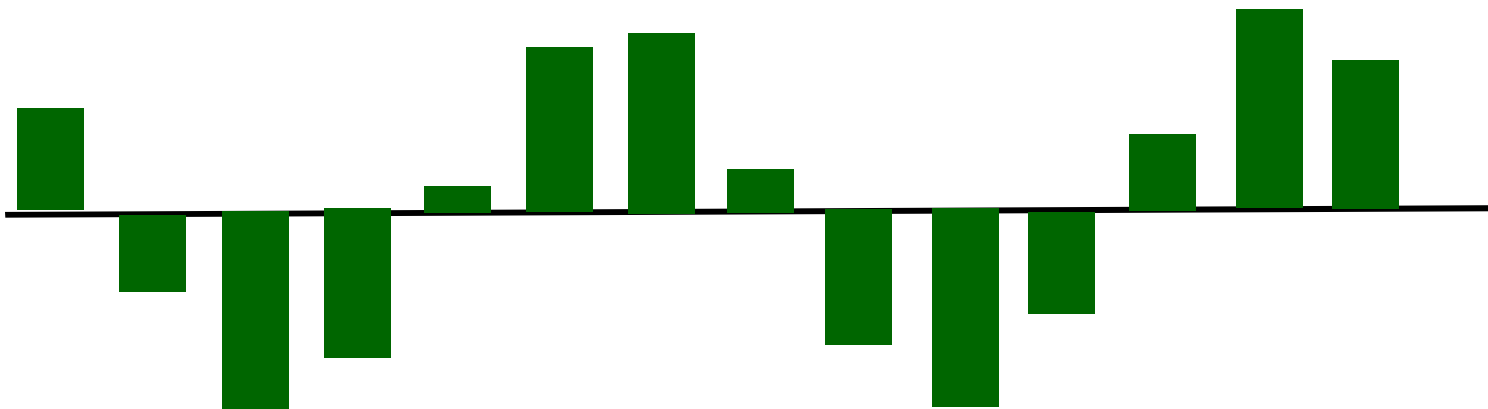


Application of Sampling Theorem

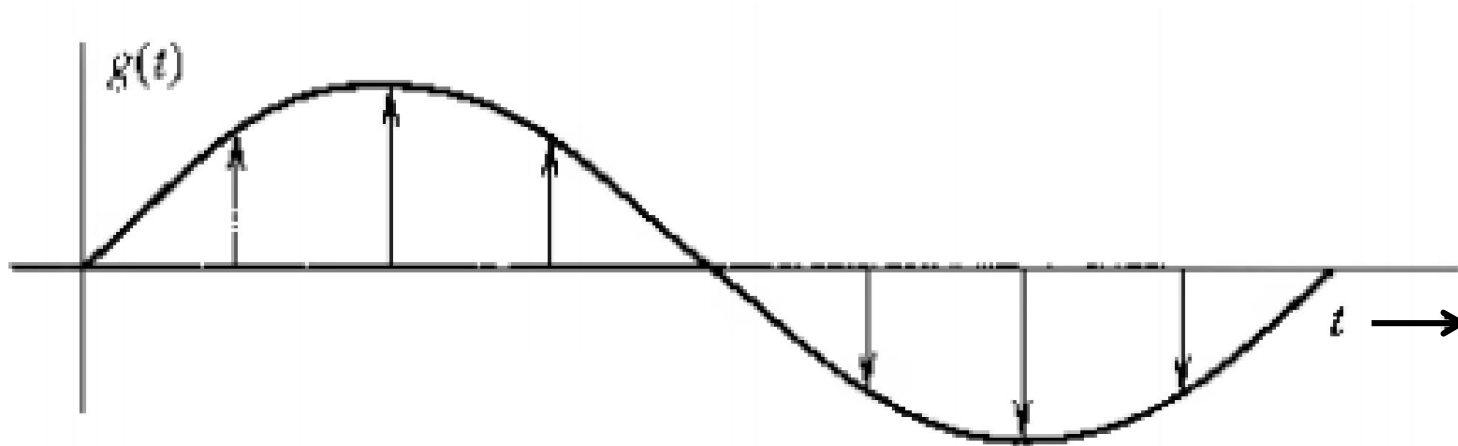
Continuous time signal



can be represented by pulse train and transmitted thereafter

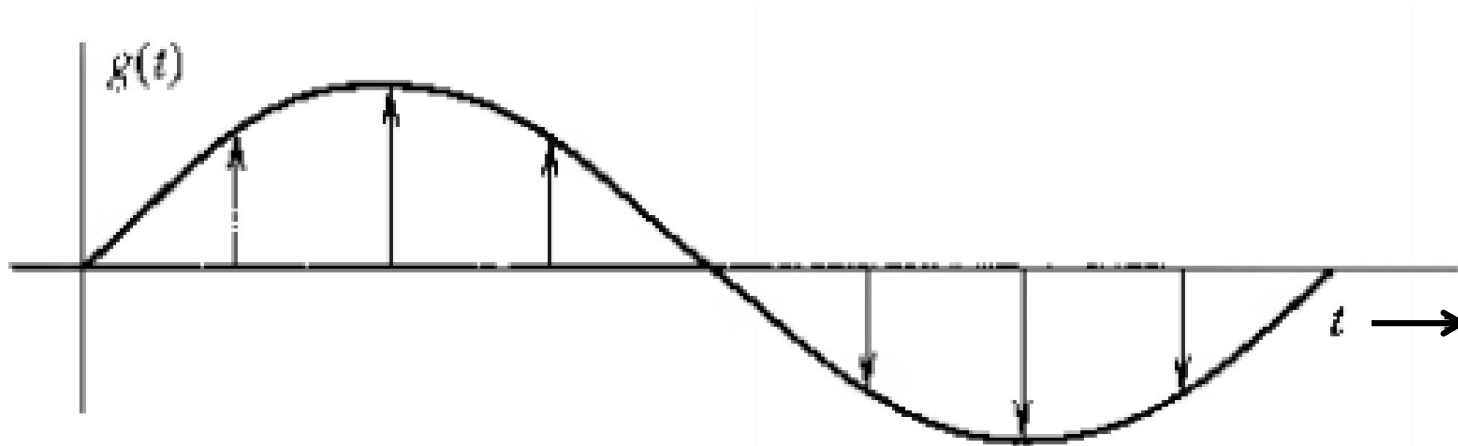


Application of Sampling Theorem

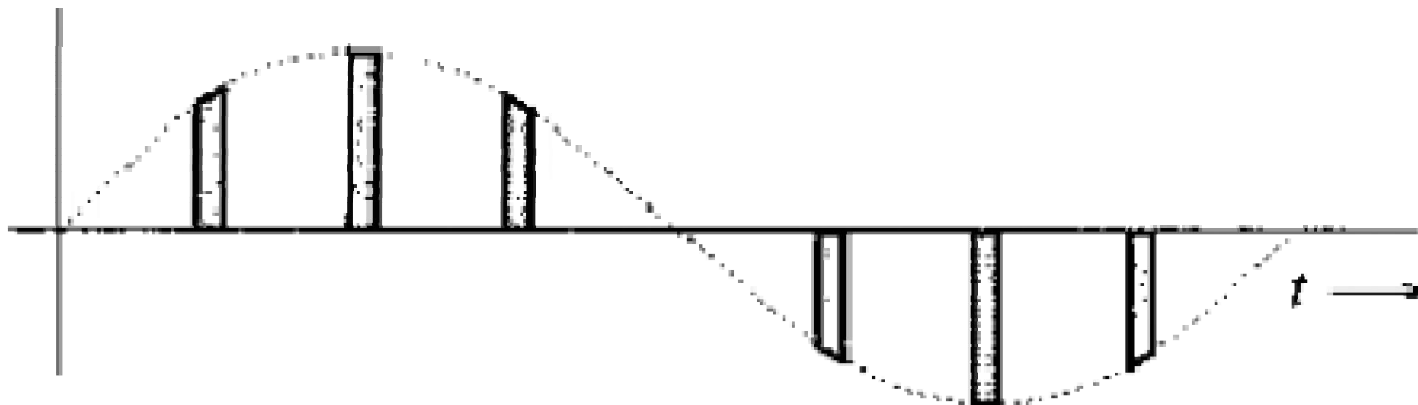


Pulse modulations: different ways to transmit sampled signal
modifying pulse trains

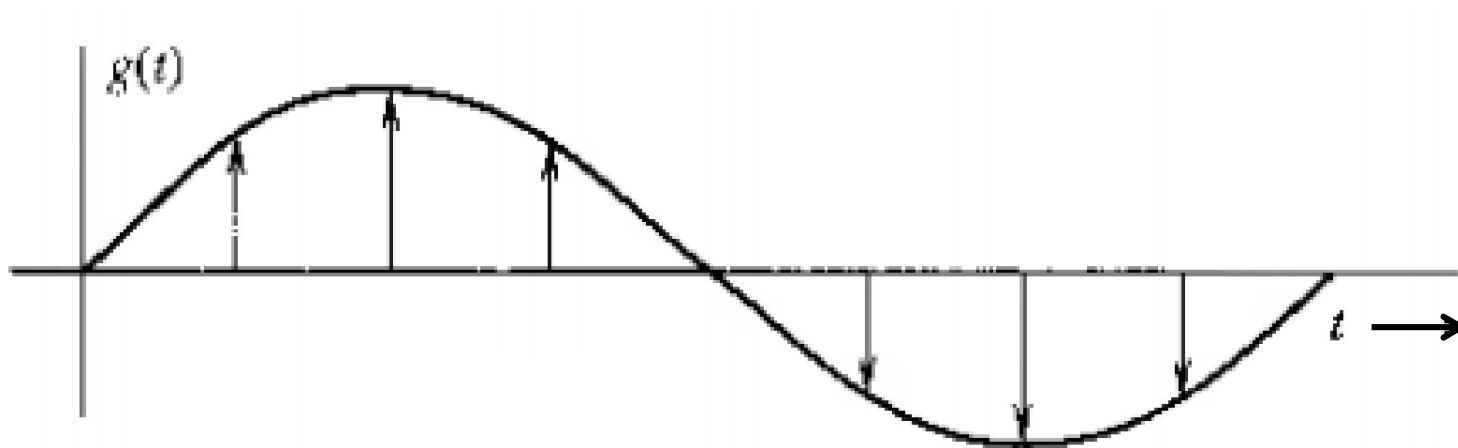
Application of Sampling Theorem



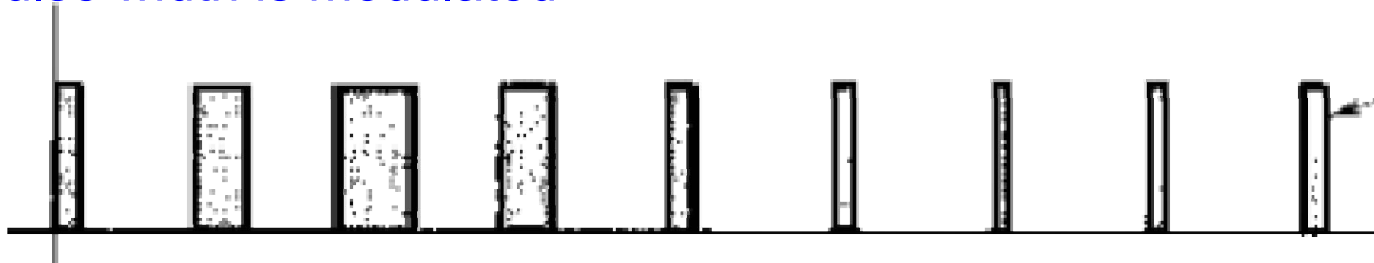
PAM: Pulse amplitude is modulated



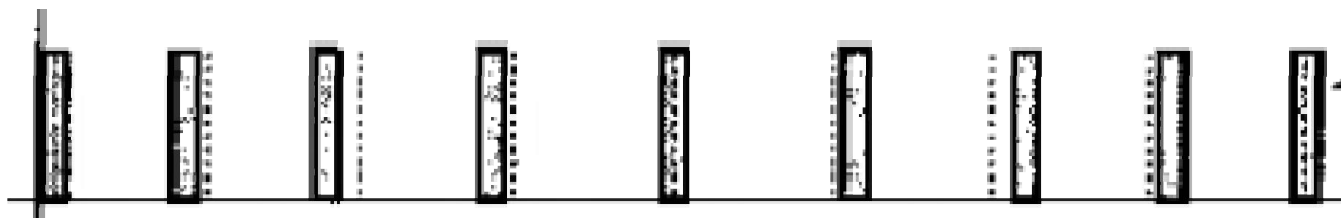
Application of Sampling Theorem



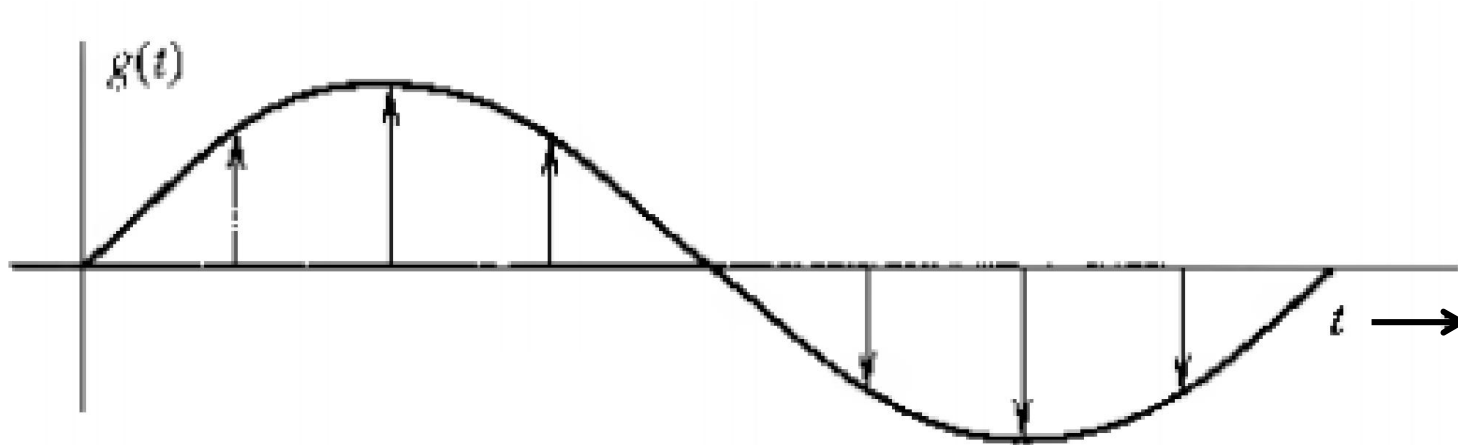
PWM: Pulse width is modulated



PPM: Pulse position is modulated



Application of Sampling Theorem

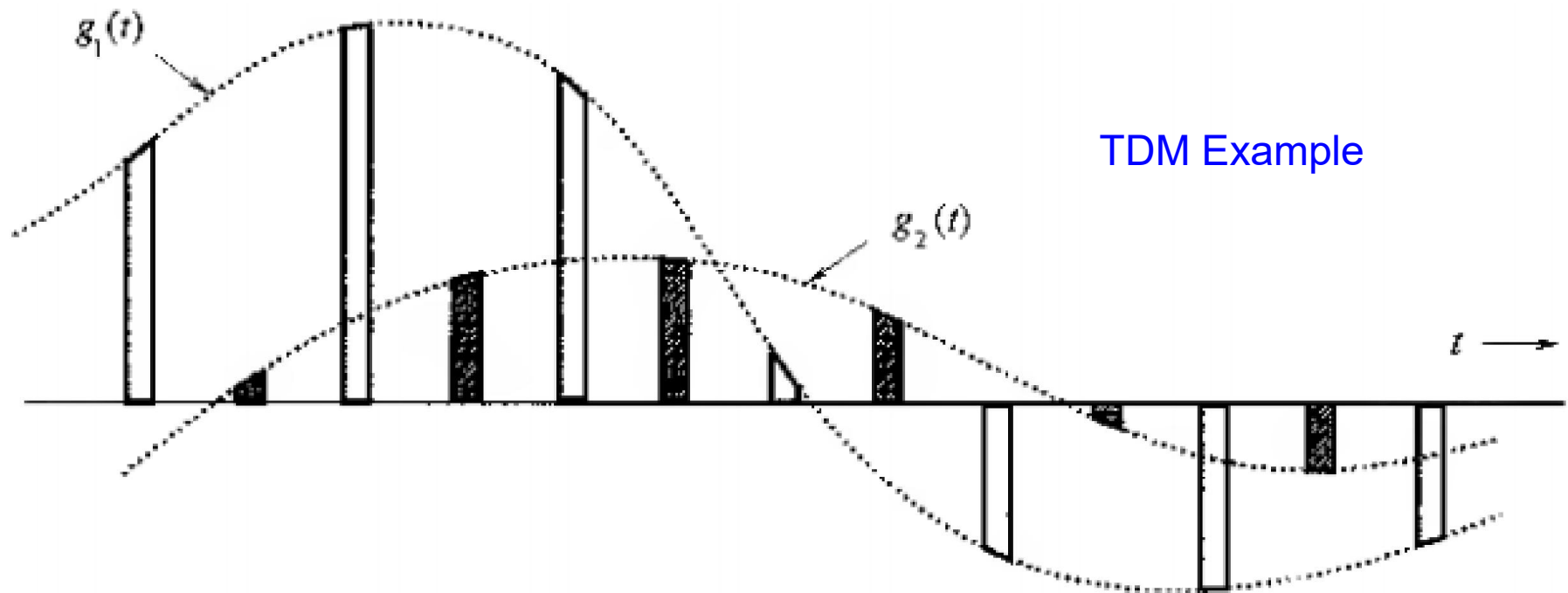


Pulse code modulation:

- most widely used pulse modulation
- each sample value is converted to a set of pulses.

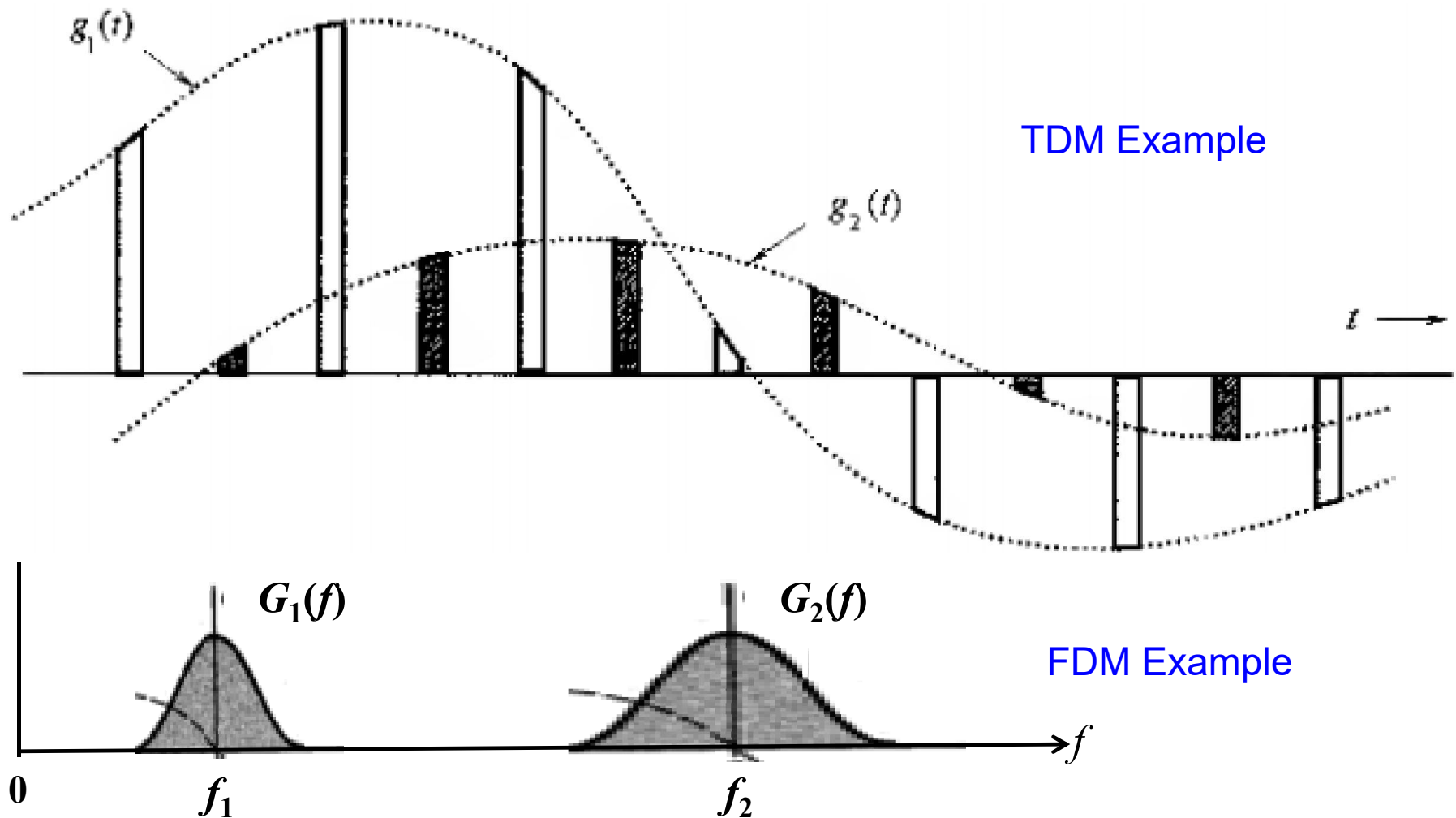
Application of Sampling Theorem

TDM: Pulses from multiple signals are interweaved on the same channel

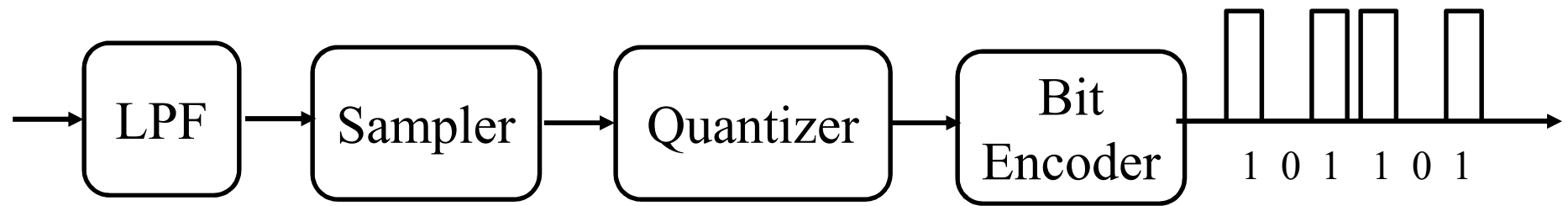


Application of Sampling Theorem

TDM: dual of FDM where different signals share channel bandwidth



Pulse Code Modulation (PCM)



PCM system: basically an ADC

Two major Steps:

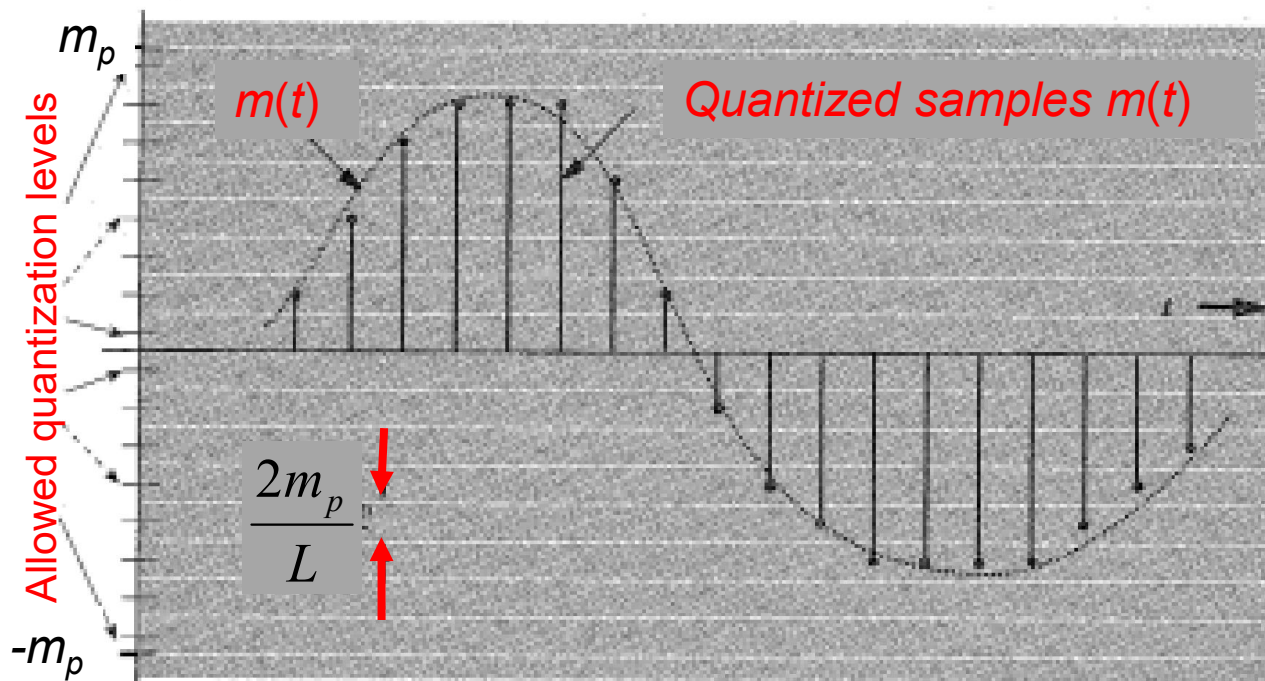
- Sampling and
- quantizing

Analog to Digital Conversion of Message Signal

- 2 major steps
 - Sampling
 - Quantizing

The range $(-m_p, m_p)$ is divided into L sub-intervals, each of magnitude Δv

$$\Delta v = \frac{2m_p}{L}$$



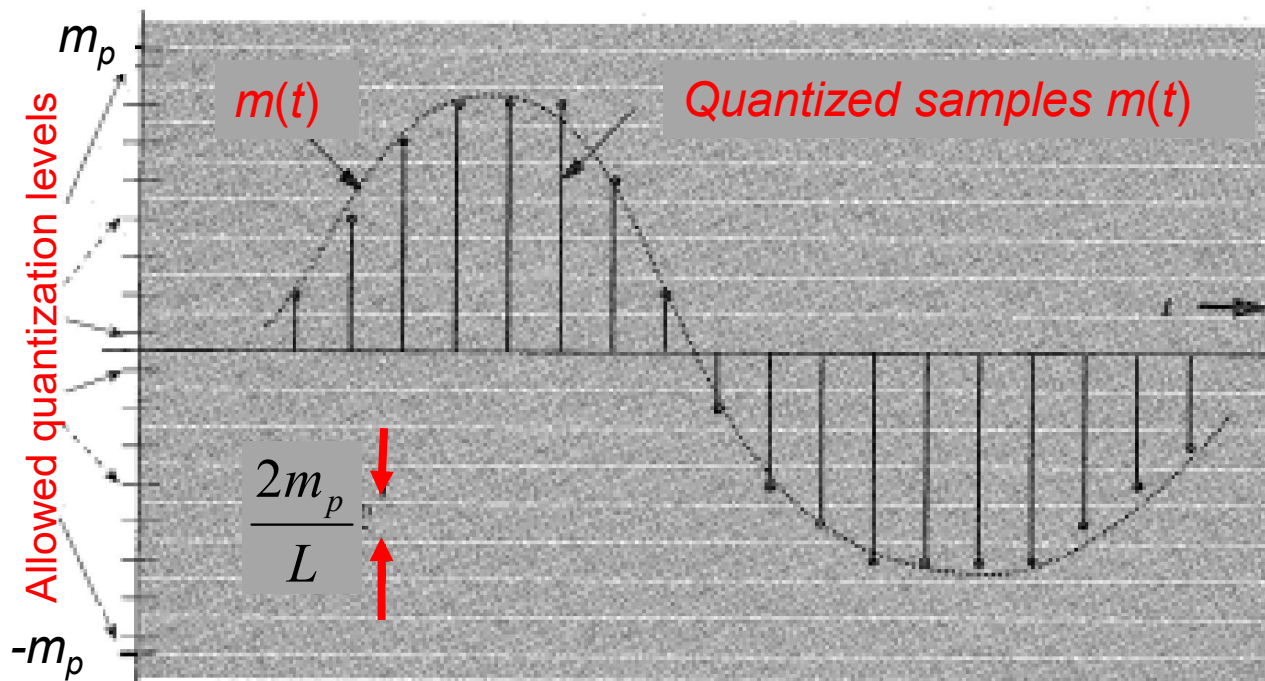
Analog to Digital Conversion of Message Signal

- 2 major steps
 - Sampling
 - Quantizing

The range $(-m_p, m_p)$ is divided into L sub-intervals, each of magnitude Δv

$$\Delta v = \frac{2m_p}{L}$$

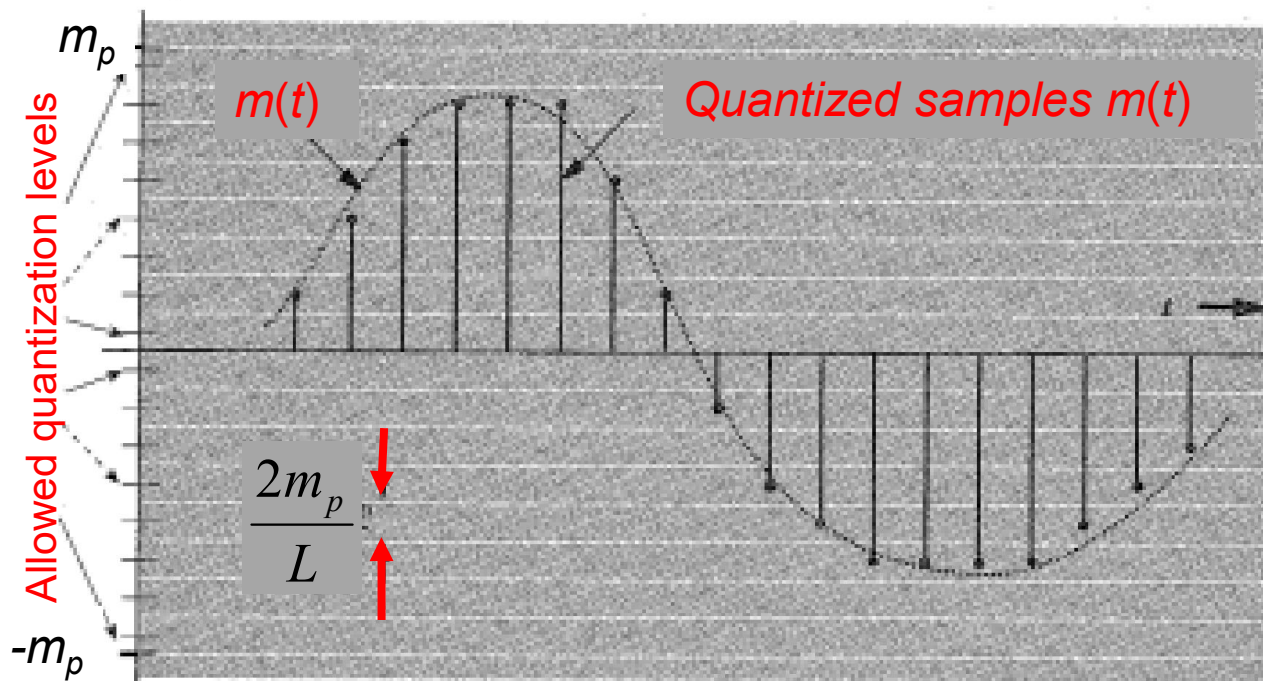
L is known as quantization level



Analog to Digital Conversion of Message Signal

- 2 major steps
 - Sampling
 - Quantizing

A **sampled value** is placed into one of these L sub-intervals, thus gets one of the L values

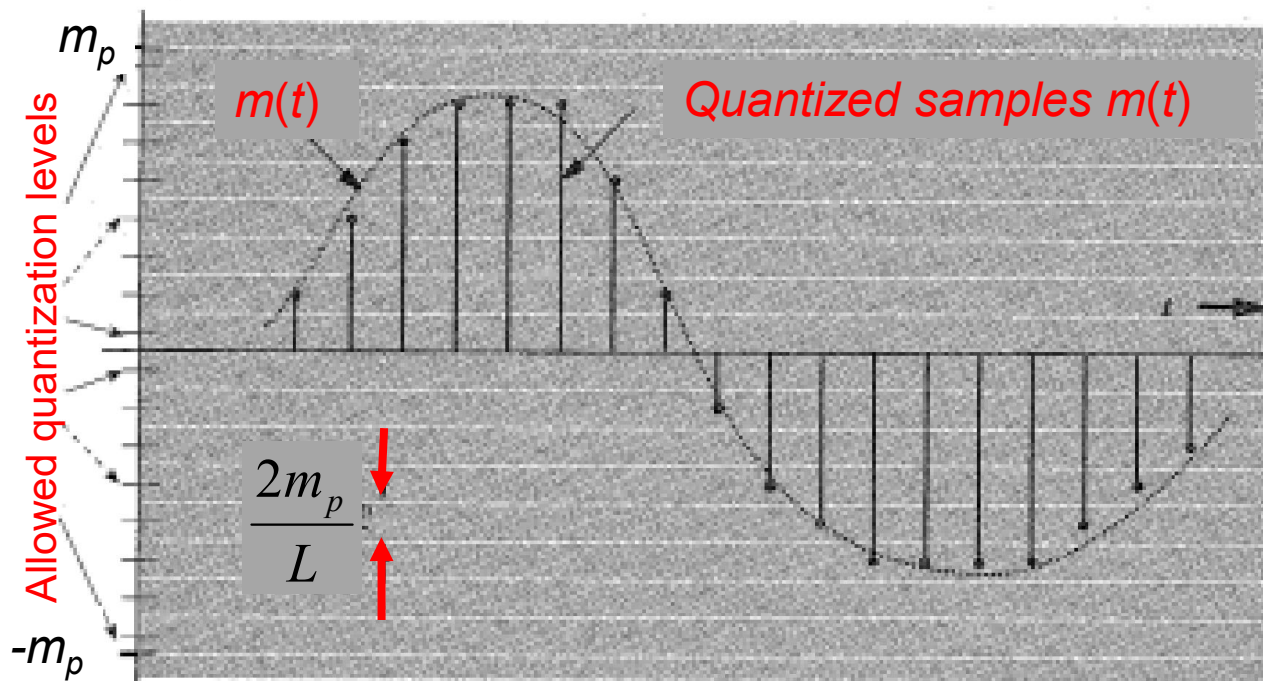


Analog to Digital Conversion of Message Signal

- 2 major steps
 - Sampling
 - Quantizing

A **sampled value** is placed into one of these L sub-intervals, thus gets one of the L values

Signal is known as L -ary digital signal



Analog to Digital Conversion of Message Signal

L -ary digital signal is converted to binary digital signal using pulse coding

Each of L values is encoded as a group of binary digits

Digit	Binary equivalent
0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001
10	1010
11	1011
12	1100
13	1101
14	1110
15	1111

Analog to Digital Conversion of Message Signal

L -ary digital signal is converted to binary digital signal using pulse coding

Each of L values is encoded as a group of binary digits

Each bit is transmitted using a distinct pulse shape

Digit	Binary equivalent	Pulse code waveform
0	0000	
1	0001	
2	0010	
3	0011	
4	0100	
5	0101	
6	0110	
7	0111	
8	1000	
9	1001	
10	1010	
11	1011	
12	1100	
13	1101	
14	1110	
15	1111	

Analog to Digital Conversion of Message Signal

Analog signal bandwidth to digital data rate

Audio signal b/w = 15 KHz

However, up to 3400 Hz is sufficient for articulation (intelligibility).

Fidelity is compromised!

Analog to Digital Conversion of Message Signal

Analog signal bandwidth to digital data rate

Audio signal b/w = 15 KHz

However, up to 3400 Hz is sufficient for articulation (intelligibility)

$$B = 3400 \text{ Hz}$$

$$f_s = 8000 > 2B$$

Analog to Digital Conversion of Message Signal

Analog signal bandwidth to digital data rate

Audio signal b/w = 15 KHz

However, up to 3400 Hz is sufficient for articulation (intelligibility)

$$B = 3400 \text{ Hz}$$

$$f_s = 8000 > 2B$$

Quantization level, $L = 256$ (8 bits)

Analog to Digital Conversion of Message Signal

Analog signal bandwidth to digital data rate

Audio signal b/w = 15 KHz

However, up to 3400 Hz is sufficient for articulation (intelligibility)

$$B = 3400 \text{ Hz}$$

$$f_s = 8000 > 2B$$

Quantization level, $L = 256$ (8 bits)

$$\text{Data rate} = 8000 * 8 = 64000 \text{ pulse/second} = 64 \text{ Kbps}$$

Analog to Digital Conversion of Message Signal

Example 2: data rate for compact disc

Fidelity is required!

Audio signal b/w = 20 KHz

$$B = 20000 \text{ Hz}$$

$$f_s = 441000 \text{ Hz} > 2B$$

Quantization level, $L = 65,536$ (16 bits)

$$\text{Data rate} = 44100 \times 16 = 1.4 \text{ Mbps}$$

Advantages of Digital Communication

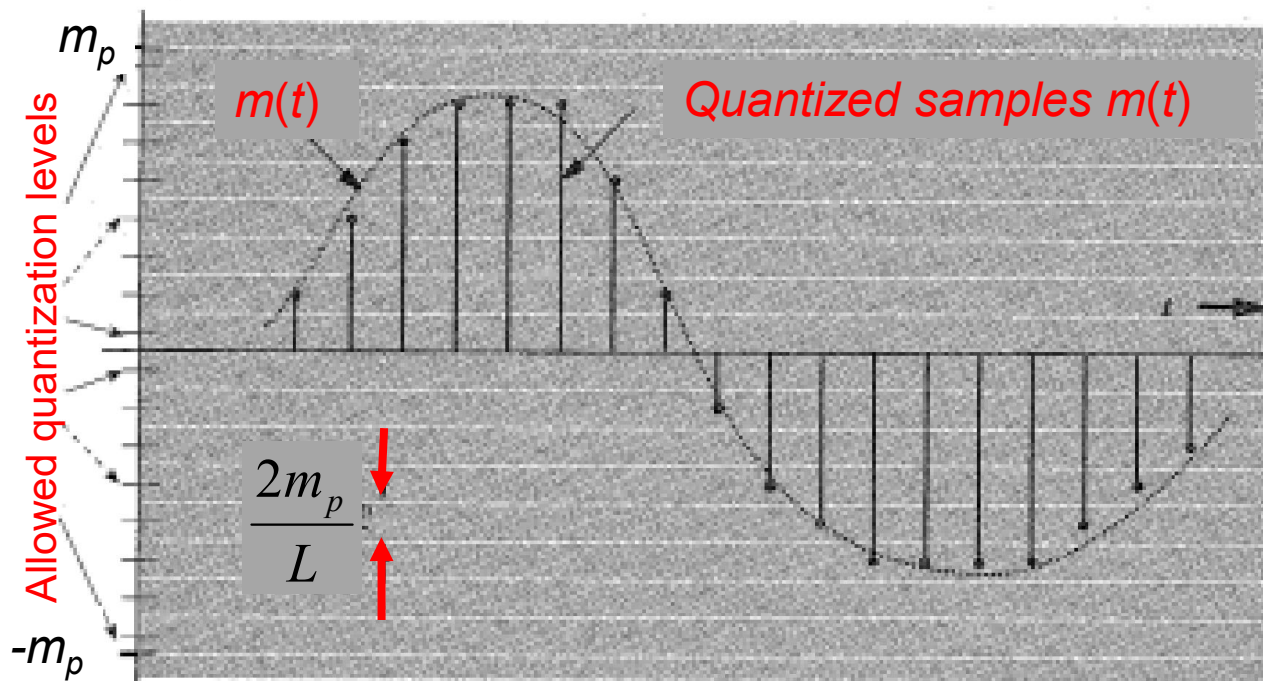
Self Study

Quantization

The range $(-m_p, m_p)$ is divided into L sub-intervals, each of magnitude Δv

$$\Delta v = \frac{2m_p}{L}$$

L is known as quantization level



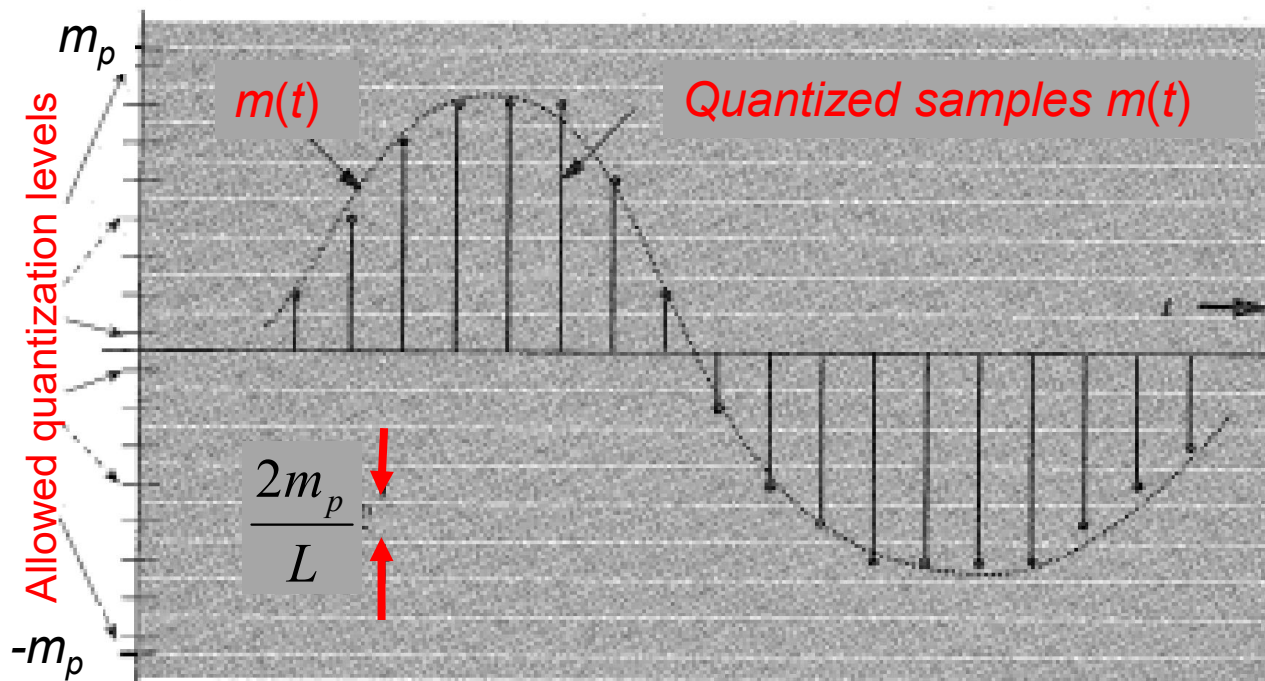
Quantization

m_p is NOT signal PEAK, rather is it's the LIMIT of the quantizer

The range $(-m_p, m_p)$ is divided into L sub-intervals, each of magnitude Δv

$$\Delta v = \frac{2m_p}{L}$$

L is known as quantization level



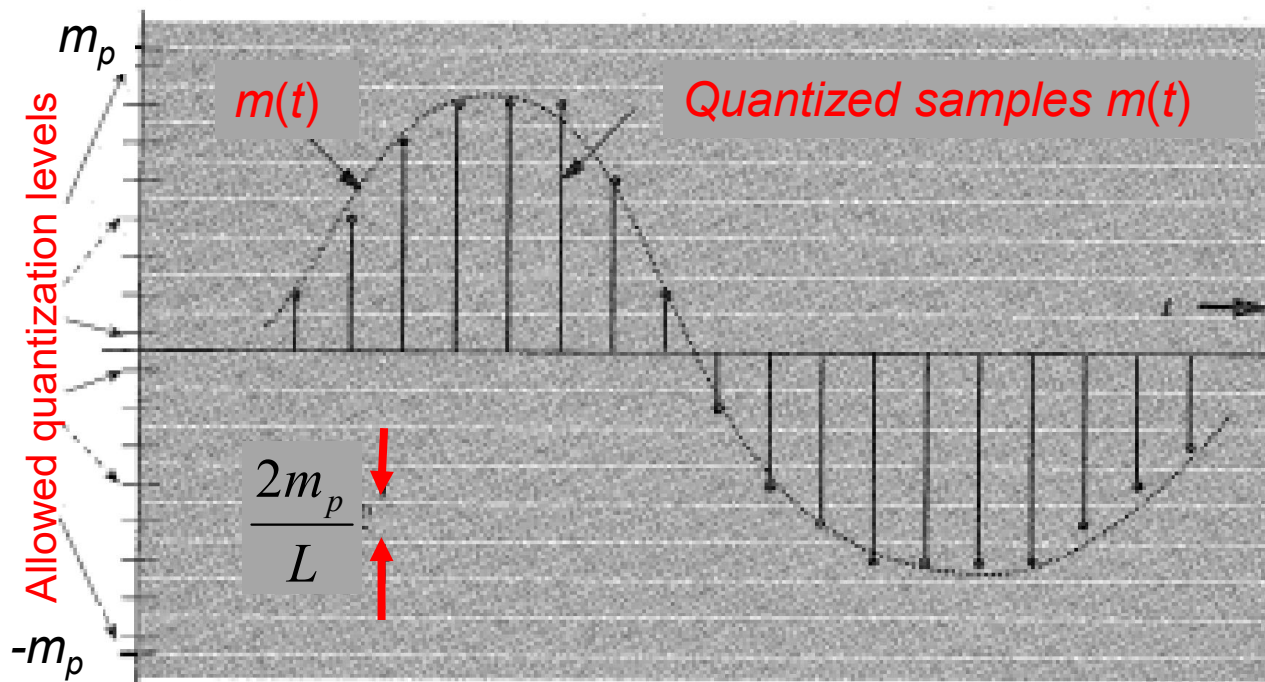
Quantization

k -th sample value $m(kT_s)$ is replaced by the midpoint of an interval where it lies

The range $(-m_p, m_p)$ is divided into L sub-intervals, each of magnitude Δv

$$\Delta v = \frac{2m_p}{L}$$

L is known as quantization level



Quantization

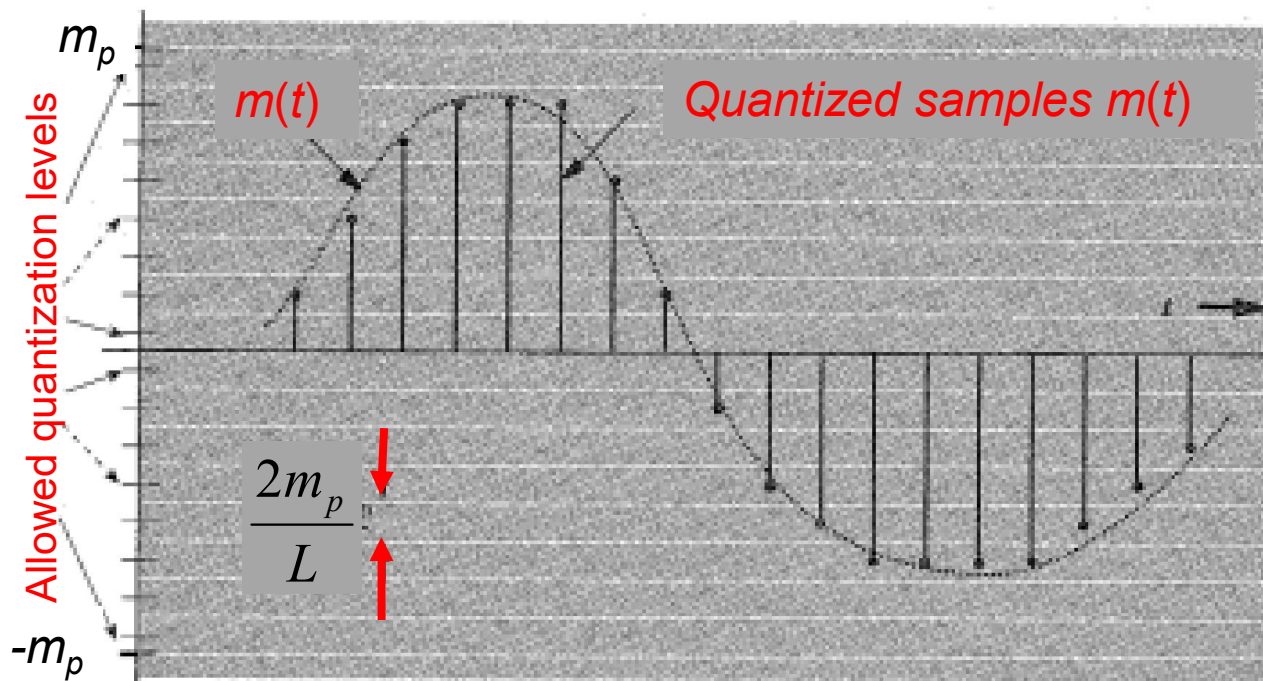
k -th sample value $m(kT_s)$ is replaced by the midpoint of an interval where it lies

$$m(kT_s) \xrightarrow{\text{Replaced by}} \hat{m}(kT_s)$$

The range $(-m_p, m_p)$ is divided into L sub-intervals, each of magnitude Δv

$$\Delta v = \frac{2m_p}{L}$$

L is known as quantization level

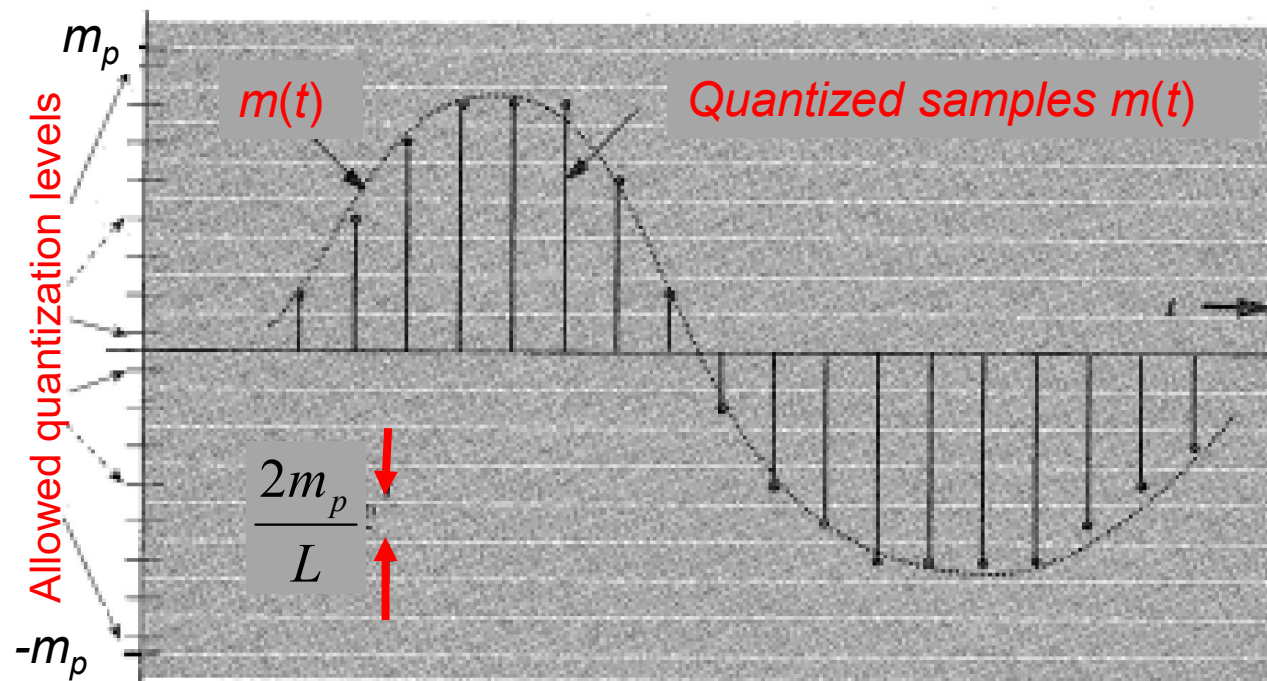


Quantization

k -th sample value $m(kT_s)$ is replaced by the midpoint of an interval where it lies

$$m(kT_s) \xrightarrow{\text{Replaced by}} \hat{m}(kT_s)$$

quantization error is unavoidable which lies in $(-\Delta v/2, \Delta v/2)$



$$\Delta v = \frac{2m_p}{L}$$

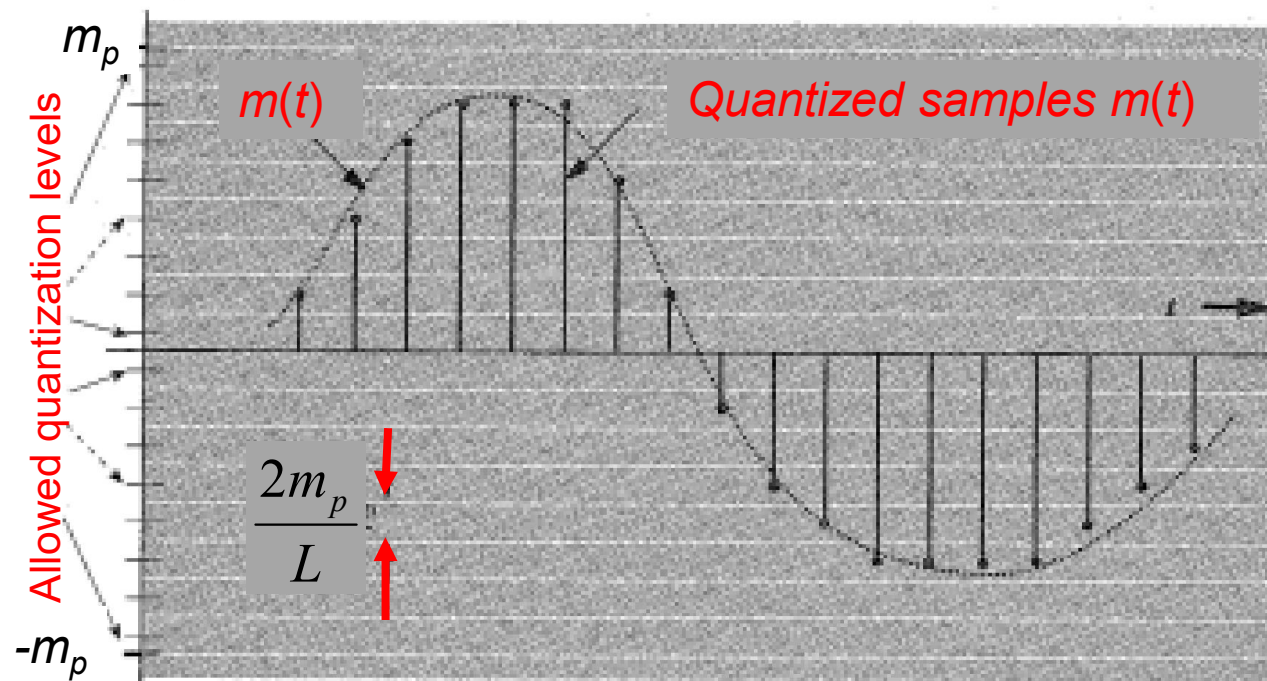
L is known as quantization level

Quantization

k -th sample value $m(kT_s)$ is replaced by the midpoint of an interval where it lies

$$m(kT_s) \xrightarrow{\text{Replaced by}} \hat{m}(kT_s)$$

quantization error is unavoidable which lies in $(-\Delta v/2, \Delta v/2)$



$$\begin{aligned} &\hat{m}(kT_s) + \Delta v / 2 \\ &\hat{m}(kT_s) \\ &\hat{m}(kT_s) - \Delta v / 2 \end{aligned}$$

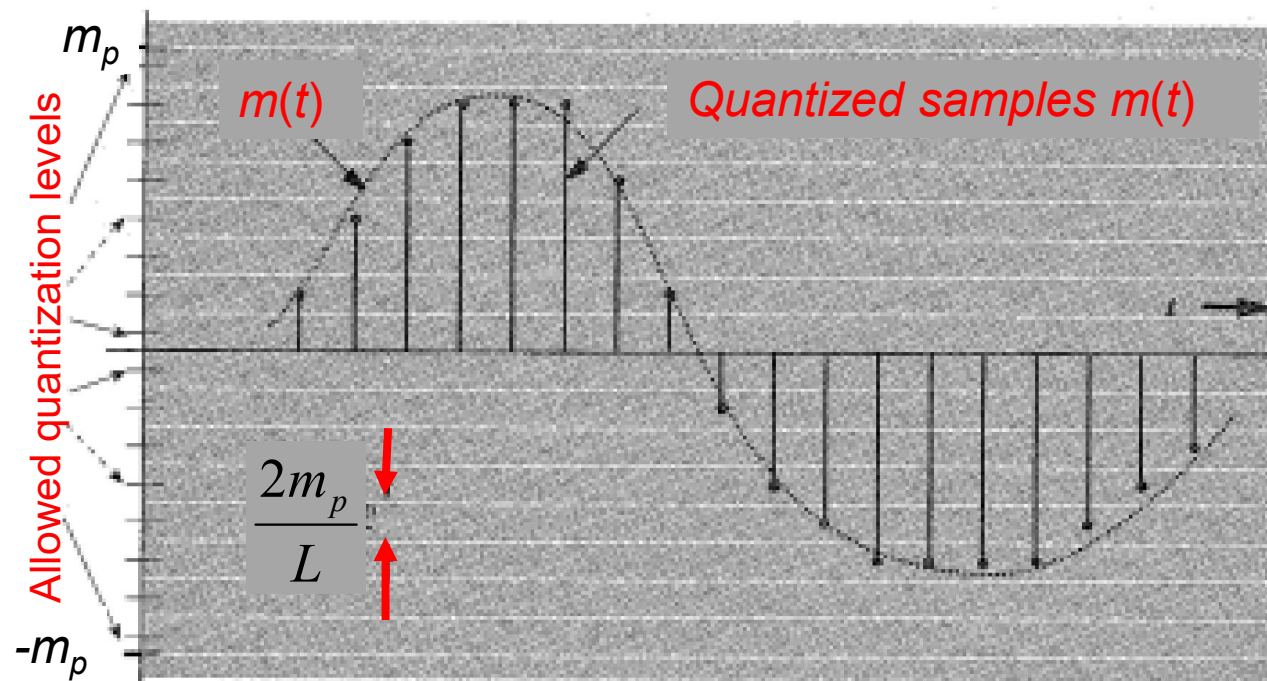
$$\Delta v = \frac{2m_p}{L}$$

Quantization

k -th sample value $m(kT_s)$ is replaced by the midpoint of an interval where it lies

$$m(kT_s) \xrightarrow{\text{Replaced by}} \hat{m}(kT_s)$$

quantization error is unavoidable which lies in $(-\Delta v/2, \Delta v/2)$



$$\begin{aligned} &\hat{m}(kT_s) + \Delta v / 2 \\ &\hat{m}(kT_s) \\ &\hat{m}(kT_s) - \Delta v / 2 \end{aligned}$$

$m(kT_s)$

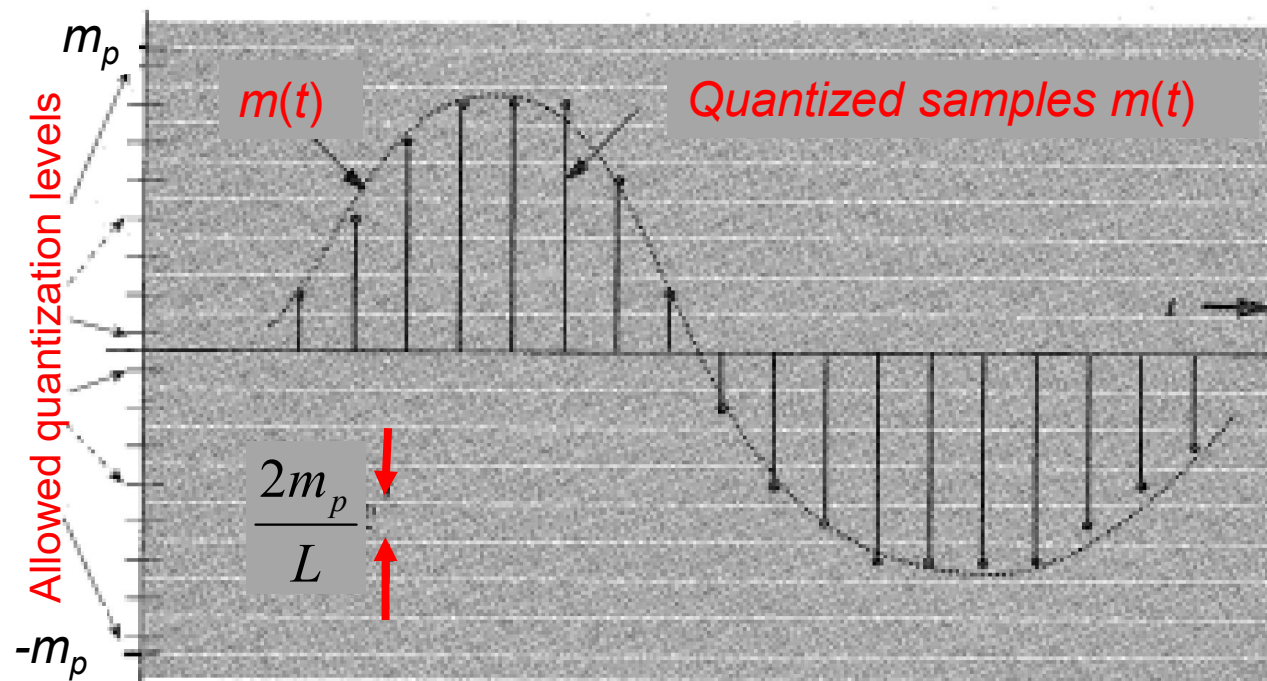
$$\Delta v = \frac{2m_p}{L}$$

Quantization

k -th sample value $m(kT_s)$ is replaced by the midpoint of an interval where it lies

$$m(kT_s) \xrightarrow{\text{Replaced by}} \hat{m}(kT_s)$$

quantization error is unavoidable which lies in $(-\Delta v/2, \Delta v/2)$



$$\begin{aligned} &\hat{m}(kT_s) + \Delta v / 2 \\ &\hat{m}(kT_s) \\ &\hat{m}(kT_s) - \Delta v / 2 \end{aligned}$$

$m(kT_s)$

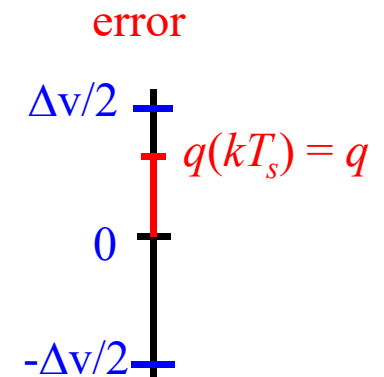
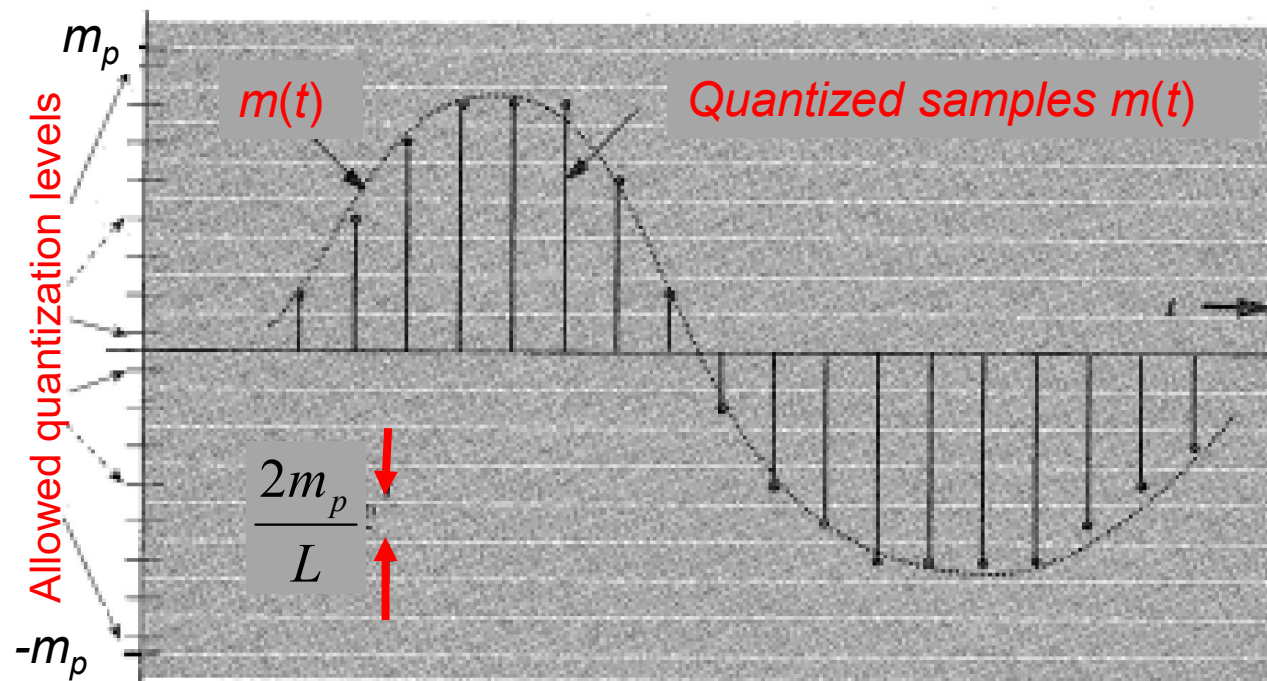
$$\Delta v = \frac{2m_p}{L}$$

Quantization

k -th sample value $m(kT_s)$ is replaced by the midpoint of an interval where it lies

$$m(kT_s) \xrightarrow{\text{Replaced by}} \hat{m}(kT_s)$$

quantization error is unavoidable which lies in $(-\Delta v/2, \Delta v/2)$



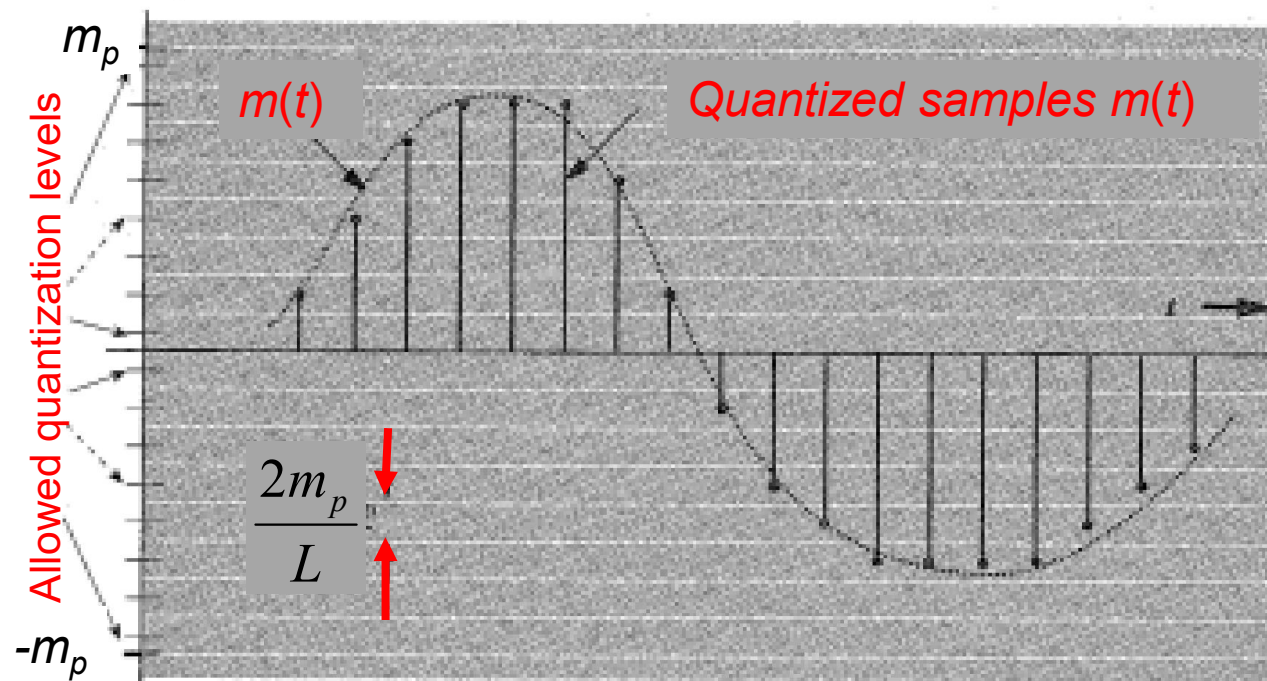
$$\Delta v = \frac{2m_p}{L}$$

Quantization

k -th sample value $m(kT_s)$ is replaced by the midpoint of an interval where it lies

$$m(kT_s) \xrightarrow{\text{Replaced by}} \hat{m}(kT_s)$$

quantization error is unavoidable which lies in $(-\Delta v/2, \Delta v/2)$



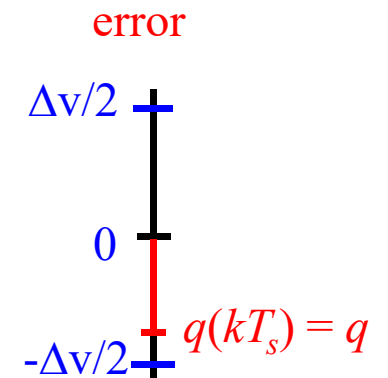
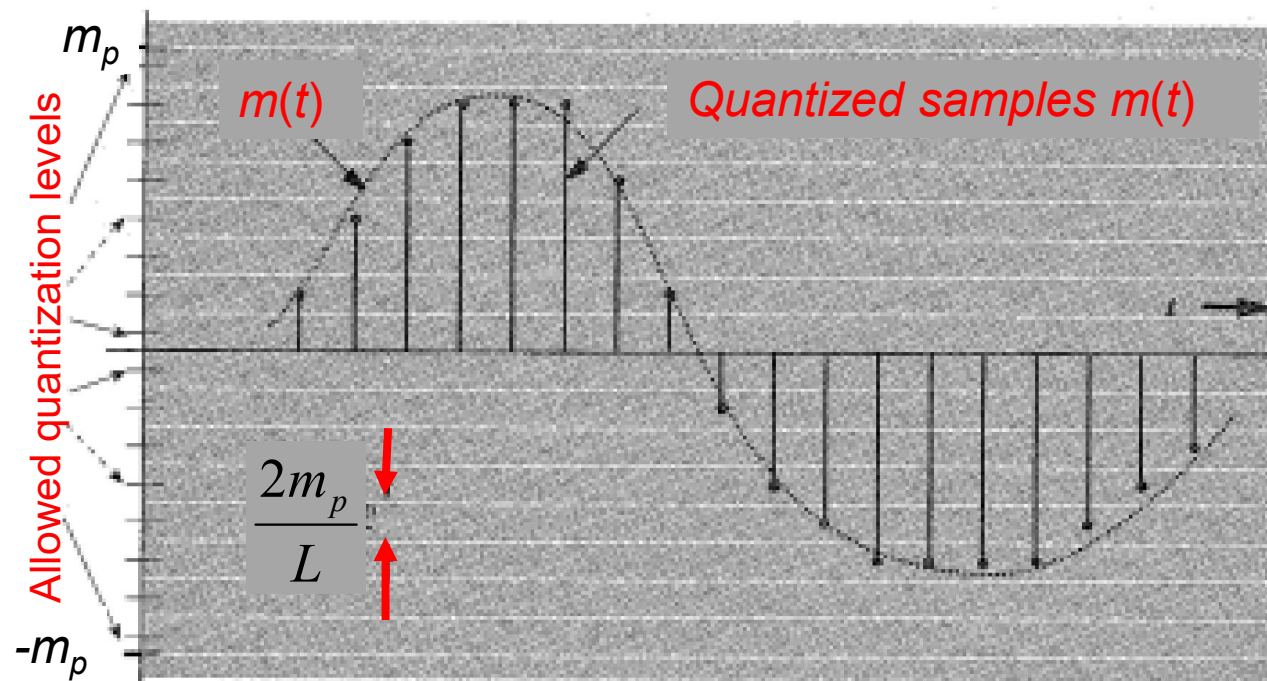
$$\Delta v = \frac{2m_p}{L}$$

Quantization

k -th sample value $m(kT_s)$ is replaced by the midpoint of an interval where it lies

$$m(kT_s) \xrightarrow{\text{Replaced by}} \hat{m}(kT_s)$$

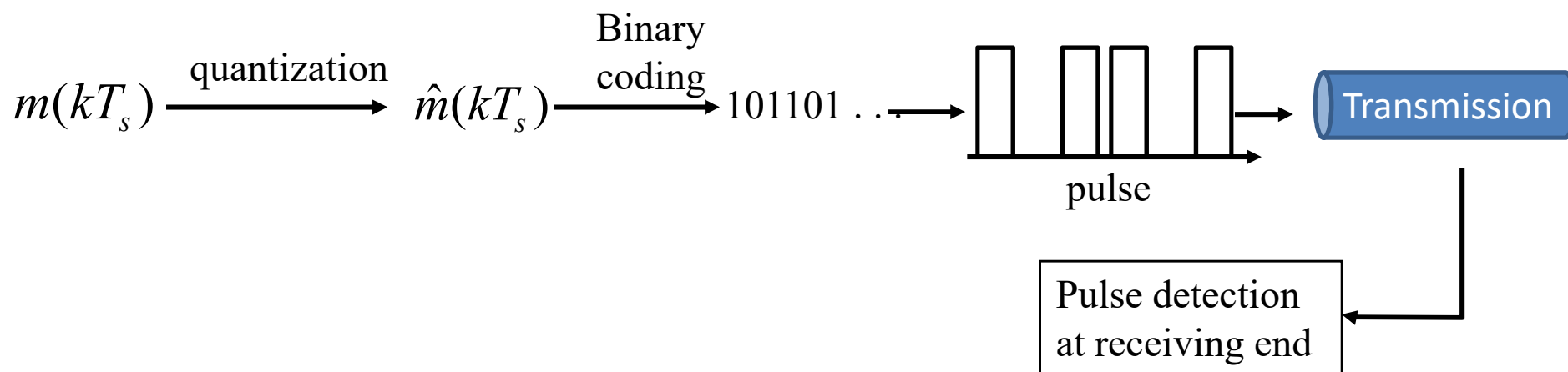
quantization error is unavoidable which lies in $(-\Delta v/2, \Delta v/2)$



$$\Delta v = \frac{2m_p}{L}$$

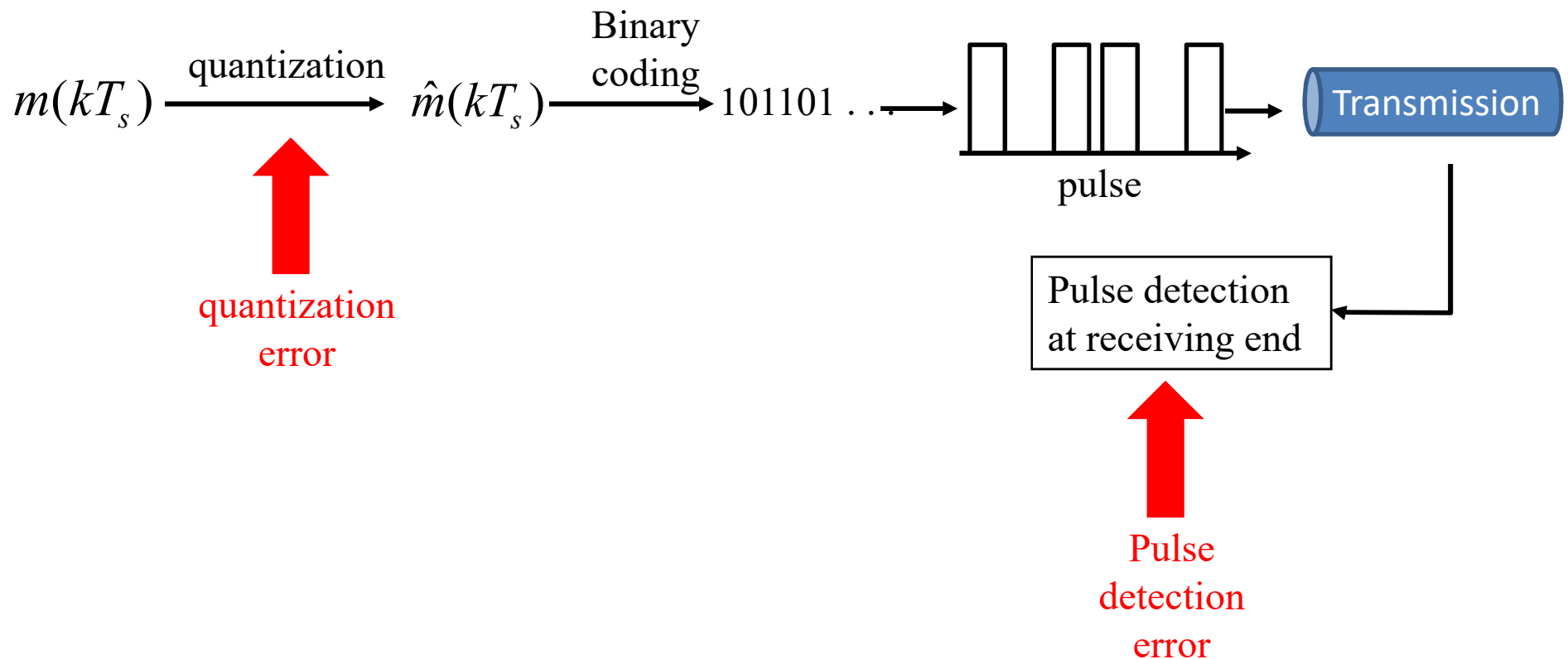
Quantization

k -th sample value $m(kT_s)$ is replaced by the midpoint of an interval where it lies



Quantization

k -th sample value $m(kT_s)$ is replaced by the midpoint of an interval where it lies



Quantization

If there were no quantization error,

$$m(t) = \sum_k m(kT_s) \text{sinc}(2\pi Bt - k\pi)$$

Quantization

If there were no quantization error,

$$m(t) = \sum_k m(kT_s) \text{sinc}(2\pi Bt - k\pi)$$

Due to quantization error,

$$\hat{m}(t) = \sum_k \hat{m}(kT_s) \text{sinc}(2\pi Bt - k\pi)$$

Quantization

If there were no quantization error,

$$m(t) = \sum_k m(kT_s) \text{sinc}(2\pi Bt - k\pi)$$

Due to quantization error,

$$\hat{m}(t) = \sum_k \hat{m}(kT_s) \text{sinc}(2\pi Bt - k\pi)$$

Quantization error $q(t)$,

$$q(t) = \hat{m}(t) - m(t)$$

Quantization

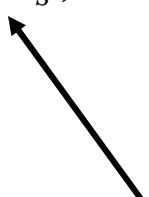
Quantization error or quantization noise or undesired signal,

$$\begin{aligned} q(t) &= \sum_k [\hat{m}(kT_s) - m(kT_s)] \text{sinc}(2\pi Bt - k\pi) \\ &= \sum_k q(kT_s) \text{sinc}(2\pi Bt - k\pi) \end{aligned}$$

Quantization

Quantization noise,

$$\begin{aligned} q(t) &= \sum_k [\hat{m}(kT_s) - m(kT_s)] \text{sinc}(2\pi Bt - k\pi) \\ &= \sum_k q(kT_s) \text{sinc}(2\pi Bt - k\pi) \end{aligned}$$



$q(kT_s)$ = Quantization
error for k th sample

Quantization

Power or Mean square of Quantization noise,

$$\begin{aligned} \overline{q^2(t)} &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} q(t)^2 dt \\ &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} \left[\sum_k q(kT_s) \text{sinc}(2\pi Bt - k\pi) \right]^2 dt \end{aligned}$$

Quantization

Power or Mean square of Quantization noise,

$$\begin{aligned}\overline{q^2(t)} &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} q(t)^2 dt \\ &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} \left[\sum_k q(kT_s) \text{sinc}(2\pi Bt - k\pi) \right]^2 dt \\ &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} [a_1 + a_2 + a_3 + \dots]^2 dt\end{aligned}$$

Quantization

Power or Mean square of Quantization noise,

$$\begin{aligned}\overline{q^2(t)} &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} q(t)^2 dt \\ &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} \left[\sum_k q(kT_s) \text{sinc}(2\pi Bt - k\pi) \right]^2 dt \\ &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} [a_1 + a_2 + a_3 + \dots]^2 dt \\ &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} [a_1^2 + a_2^2 + a_3^2 + \dots + 2a_1a_2 + 2a_1a_3 + 2a_1a_4 + \dots] dt\end{aligned}$$

Quantization

Power or Mean square of Quantization noise,

$$\begin{aligned}\overline{q^2(t)} &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} q(t)^2 dt \\ &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} \left[\sum_k q(kT_s) \text{sinc}(2\pi Bt - k\pi) \right]^2 dt \\ &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} \left[a_1^2 + a_2^2 + a_3^2 + \cdots + 2a_1a_2 + 2a_1a_3 + 2a_1a_4 + \cdots \right] dt \\ &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} \left[\sum_k a_k^2 + 2 \sum_{m \neq n} a_m a_n \right] dt\end{aligned}$$

Quantization

Power or Mean square of Quantization noise,

$$\begin{aligned}\overline{q^2(t)} &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} \left[\sum_k q(kT_s) \text{sinc}(2\pi Bt - k\pi) \right]^2 dt \\ &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} \left[\sum_k a_k^2 + 2 \sum_{m \neq n} a_m a_n \right] dt \\ &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} \left[\sum_k q^2(kT_s) \text{sinc}^2(2\pi Bt - k\pi) \right] dt \\ &\quad + \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} \left[2 \sum_{m \neq n} q(mT_s) q(nT_s) \text{sinc}(2\pi Bt - m\pi) \text{sinc}(2\pi Bt - n\pi) \right] dt\end{aligned}$$

Quantization

Power or Mean square of Quantization noise,

$$\begin{aligned}
 \overline{q^2(t)} &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} \left[\sum_k q^2(kT_s) \text{sinc}^2(2\pi Bt - k\pi) \right] dt \\
 &+ \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} \left[2 \sum_{m \neq n} q(mT_s) q(nT_s) \text{sinc}(2\pi Bt - m\pi) \text{sinc}(2\pi Bt - n\pi) \right] dt \\
 &= \lim_{T \rightarrow \infty} \frac{1}{T} \sum_k q^2(kT_s) \int_{-T/2}^{T/2} \text{sinc}^2(2\pi Bt - k\pi) dt \\
 &+ \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{m \neq n} q(mT_s) q(nT_s) \int_{-T/2}^{T/2} \text{sinc}(2\pi Bt - m\pi) \text{sinc}(2\pi Bt - n\pi) dt
 \end{aligned}$$

Quantization

Power or Mean square of Quantization noise,

$$\begin{aligned} \overline{q^2(t)} &= \lim_{T \rightarrow \infty} \frac{1}{T} \sum_k q^2(kT_s) \int_{-T/2}^{T/2} \text{sinc}^2(2\pi Bt - k\pi) dt \\ &+ \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{m \neq n} q(mT_s) q(nT_s) \int_{-T/2}^{T/2} \text{sinc}(2\pi Bt - m\pi) \text{sinc}(2\pi Bt - n\pi) dt \end{aligned}$$

Quantization

Power or Mean square of Quantization noise,

$$\begin{aligned} \overline{q^2(t)} &= \lim_{T \rightarrow \infty} \frac{1}{T} \sum_k q^2(kT_s) \int_{-T/2}^{T/2} \text{sinc}^2(2\pi Bt - k\pi) dt \\ &+ \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{m \neq n} q(mT_s) q(nT_s) \int_{-T/2}^{T/2} \text{sinc}(2\pi Bt - m\pi) \text{sinc}(2\pi Bt - n\pi) dt \end{aligned}$$

We can prove that,

$$\int_{-\infty}^{\infty} \text{sinc}(2\pi Bt - m\pi) \text{sinc}(2\pi Bt - n\pi) dt = \begin{cases} 0 & m \neq n \\ \frac{1}{2B} & m = n \end{cases}$$

Quantization

Power or Mean square of Quantization noise,

$$\overline{q^2(t)} = \lim_{T \rightarrow \infty} \frac{1}{2BT} \sum_k q^2(kT_s)$$

Quantization

Power or Mean square of Quantization noise,

$$\overline{q^2(t)} = \lim_{T \rightarrow \infty} \frac{1}{2BT} \sum_k q^2(kT_s)$$

As sampling frequency $f_s = 2B$,

$2BT = \text{total no. of samples over averaging time } T$

RHS is the mean of the square of quantization error

Quantization

Power or Mean square of Quantization noise,

$$\overline{q^2(t)} = \lim_{T \rightarrow \infty} \frac{1}{2BT} \sum_k q^2(kT_s)$$

As sampling frequency $f_s = 2B$,

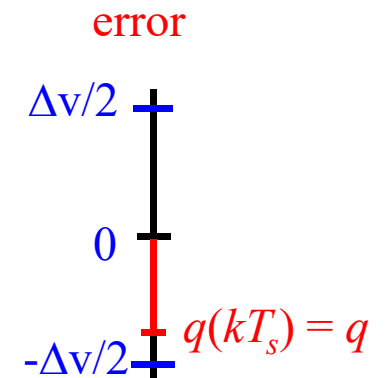
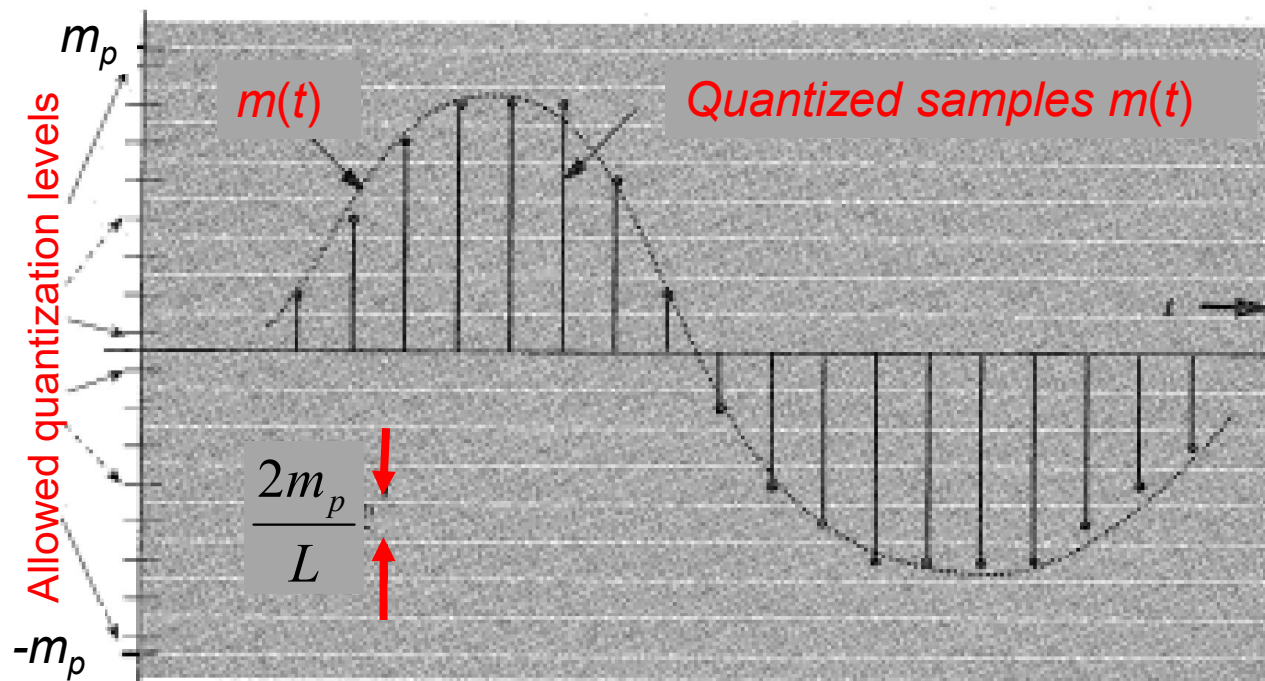
$2BT = \text{total no. of samples over averaging time } T$

RHS is the mean of the square of quantization error

Therefore, power of quantization noise = mean square quantization error

Quantization

We know, quantization error q lies in $(-\Delta v/2, \Delta v/2)$

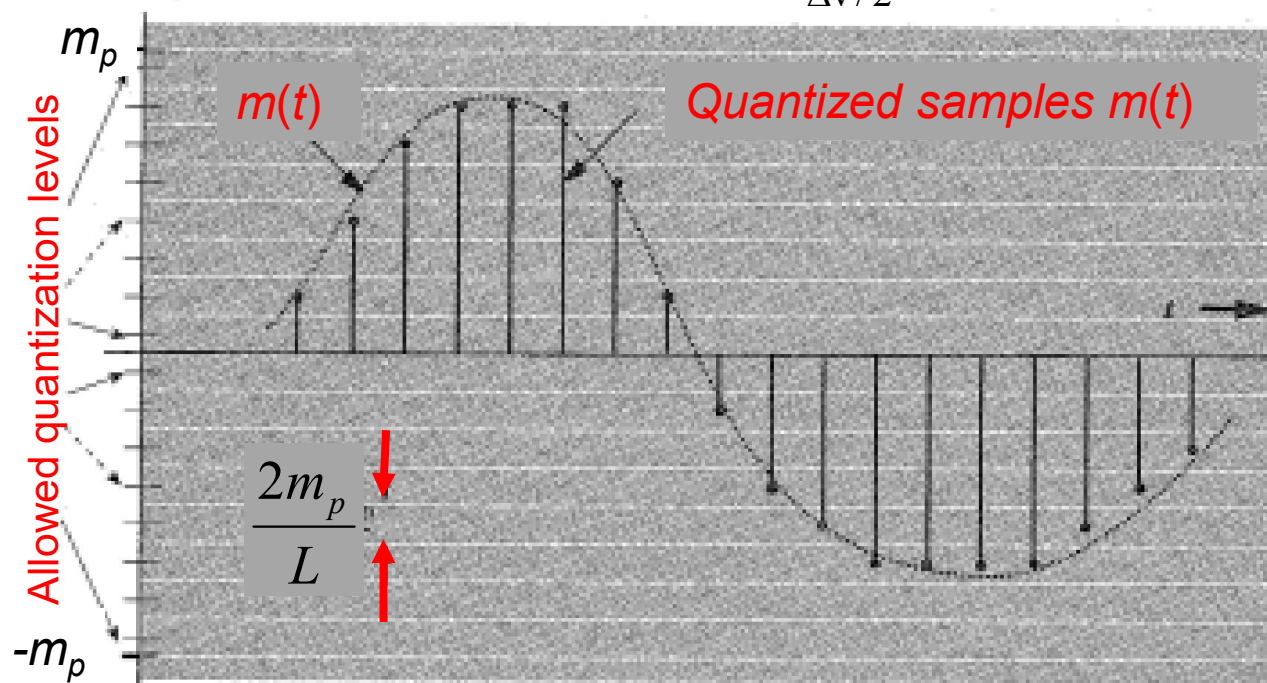


$$\Delta v = \frac{2m_p}{L}$$

Quantization

Mean square quantization error is given by

$$\overline{q^2} = \frac{1}{\Delta v} \int_{-\Delta v/2}^{\Delta v/2} q^2 dq$$



Quantization

Mean square quantization error is given by

$$\tilde{\tilde{q}}^2 = \frac{1}{\Delta v} \int_{-\Delta v/2}^{\Delta v/2} q^2 dq = \frac{(\Delta v)^2}{12}$$

Quantization

Mean square quantization error is given by

$$\overline{\overline{q^2}} = \frac{1}{\Delta v} \int_{-\Delta v/2}^{\Delta v/2} q^2 dq = \frac{(\Delta v)^2}{12} = \frac{m_p^2}{3L^2}$$

$$\text{where, } \Delta v = \frac{2m_p}{L}$$

Quantization

Mean square quantization error is given by

$$\tilde{\tilde{q}}^2 = \frac{1}{\Delta v} \int_{-\Delta v/2}^{\Delta v/2} q^2 dq = \frac{(\Delta v)^2}{12} = \frac{m_p^2}{3L^2}$$

We proved, power of quantization noise (N_0) = mean square quantization error

Quantization

Mean square quantization error is given by

$$\overline{q^2} = \frac{1}{\Delta v} \int_{-\Delta v/2}^{\Delta v/2} q^2 dq = \frac{(\Delta v)^2}{12} = \frac{m_p^2}{3L^2}$$

We proved, power of quantization noise (N_0) = mean square quantization error

$$N_0 = \overline{q^2(t)} = \overline{q^2} = \frac{m_p^2}{3L^2}$$

Quantization

power of quantization noise (N_0) = mean square quantization error

$$N_0 = \overline{q^2(t)} = \overline{q^2} = \frac{m_p^2}{3L^2}$$

Assume, power of message signal (S_0) is given by $S_0 = \overline{m^2(t)}$

Quantization

power of quantization noise (N_0) = mean square quantization error

$$N_0 = \overline{q^2(t)} = \overline{q^2} = \frac{m_p^2}{3L^2}$$

Assume, power of message signal (S_0) is given by $S_0 = \overline{m^2(t)}$

Signal-to-noise ratio (SNR) is

$$SNR = \frac{S_0}{N_0} = 3L^2 \frac{\overline{m^2(t)}}{m_p^2}$$

Quantization

$$SNR = \frac{S_0}{N_0} = 3L^2 \frac{\overline{m^2(t)}}{m_p^2}$$

- Higher SNR means higher quality of received signal
- L increases SNR
- Higher limit of quantizer (m_p) decreases SNR

Quantization

$$SNR = \frac{S_0}{N_0} = 3L^2 \frac{\overbrace{m^2(t)}}{m_p^2}$$

- Higher SNR means higher quality of received signal
- L increases SNR
- Higher limit of quantizer (m_p) decreases SNR
- SNR is linear function of signal power, $S_0 = \overbrace{m^2(t)}$

Quantization

$$SNR = \frac{S_0}{N_0} = 3L^2 \frac{\overline{m^2(t)}}{m_p^2}$$

- SNR is linear function of signal power, $S_0 = \overline{m^2(t)}$

S_0 varies

- from speaker to speaker
- due to different length of connecting circuits

Quantization

$$SNR = \frac{S_0}{N_0} = 3L^2 \frac{\overline{m^2(t)}}{m_p^2}$$

- SNR is linear function of signal power, $S_0 = \overline{m^2(t)}$

S_0 varies

- from speaker to speaker
- due to different length of connecting circuits

For these reasons,

- SNR varies widely
- Quality of received signal deteriorates remarkably for soft speakers

Quantization

$$SNR = \frac{S_0}{N_0} = 3L^2 \frac{\overline{m^2(t)}}{m_p^2}$$

- SNR is linear function of signal power, $S_0 = \overline{m^2(t)}$

S_0 varies

- from speaker to speaker
- due to different length of connecting circuits

For these reasons,

- SNR varies widely
- Quality of received signal deteriorates remarkably for soft speakers

However statistically,

- Small amplitudes (soft speakers) predominate in speech
- Larger amplitudes (loud speakers) are less frequent

Quantization

$$SNR = \frac{S_0}{N_0} = 3L^2 \frac{\overline{m^2(t)}}{m_p^2}$$

Causes of the problem

- All amplitudes are **uniformly** quantized with **equal step size** $\Delta v = \frac{2m_p}{L}$
- **Variation in amplitudes is NOT considered**

Quantization

$$SNR = \frac{S_0}{N_0} = 3L^2 \frac{\overline{m^2(t)}}{m_p^2}$$

Causes of the problem

- All amplitudes are **uniformly** quantized with **equal step size** $\Delta v = \frac{2m_p}{L}$
- Variation in amplitudes is **NOT** considered

Quantization noise $N_0 = (\Delta v)^2/12$ is directly **proportional to square of step size** Δv

$$N_0 \propto (\Delta v)^2$$

Quantization

Solution to the problem: Non-uniform quantization

- Smaller step size for smaller amplitudes
- Larger step size for larger amplitudes